

```
In [2]: import yfinance as yf
import matplotlib.pyplot as plt
import pandas as pd
from datetime import datetime, timedelta
```

```
In [3]: techStocks = {
    'AAPL': 'Apple',
    'PLTR': 'Palantir',
    'GOOGL': 'Google',
    'AMZN': 'Amazon',
    'TSLA': 'Tesla',
    'NVDA': 'NVIDIA',
    'NFLX': 'Netflix',
    'META': 'Meta',
    'MSFT': 'Microsoft'
}

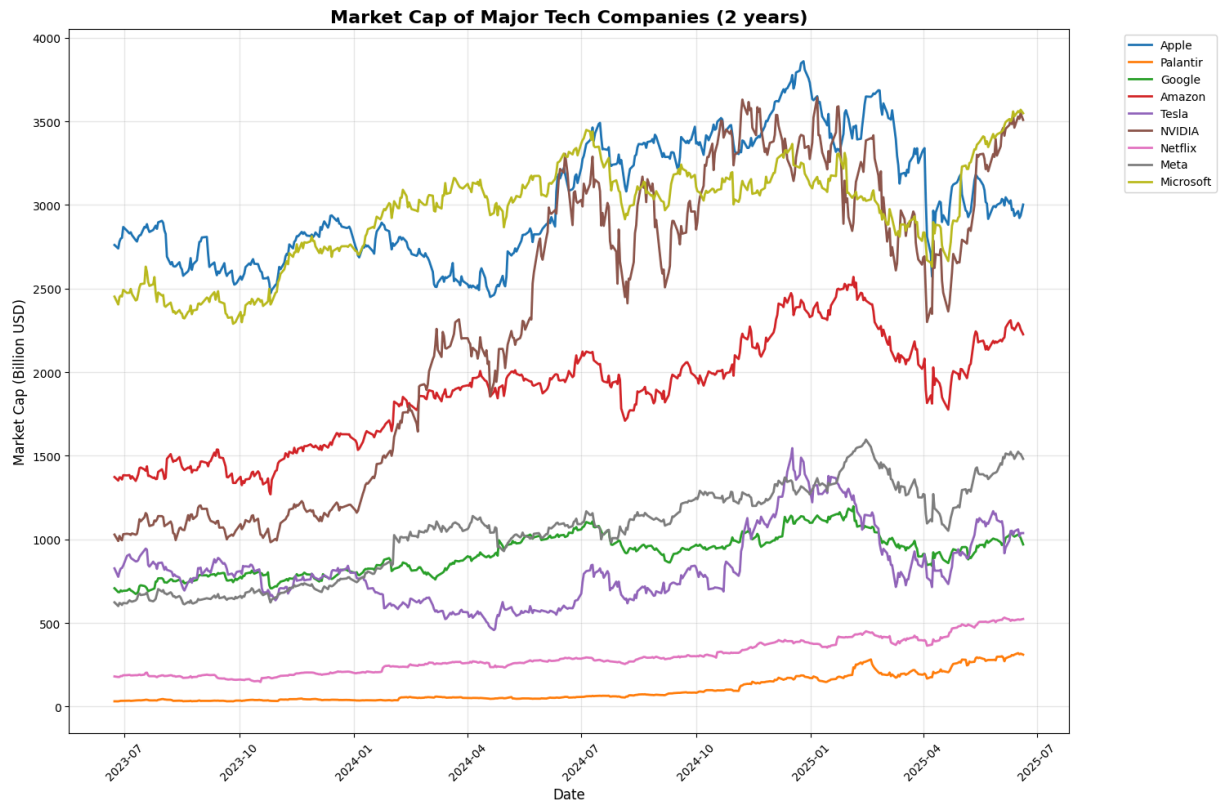
end_date = datetime.now()
start_date = end_date - timedelta(days=730)

stockData = {}
for symbol, name in techStocks.items():
    ticker = yf.Ticker(symbol)
    data = ticker.history(start=start_date, end=end_date)
    shareCount = ticker.info.get('sharesOutstanding')
    marketVal = data['Close'] * shareCount
    stockData[name] = marketVal

plt.figure(figsize=(15,10))
for name, prices in stockData.items():
    if len(prices) > 0:
        plt.plot(prices.index, prices.values / 1e9, linewidth=2, label=name)

plt.title('Market Cap of Major Tech Companies (2 years)', fontsize=16, fontweight='bold')
plt.xlabel("Date", fontsize=12)
plt.ylabel("Market Cap (Billion USD)", fontsize=12)
plt.legend(bbox_to_anchor=(1.05, 1), loc='upper left')
plt.grid(True, alpha=0.3)
plt.xticks(rotation=45)
plt.tight_layout()
plt.show()

print("\nEstimated Current Market Capitalizations:")
print("-" * 50)
for symbol, name in techStocks.items():
    ticker = yf.Ticker(symbol)
    currentPrice = ticker.history(period="1d")['Close'].iloc[-1]
    shares = ticker.info.get('sharesOutstanding')
    marketCap = currentPrice * shares
    print(f"{name:12} ({symbol}): ${marketCap / 1e9:.2f} Billion")
```



Estimated Current Market Capitalizations:

Apple (AAPL): \$3002.10 Billion

HTTP Error 401:

Palantir (PLTR): \$310.70 Billion

Google (GOOGL): \$969.84 Billion

Amazon (AMZN): \$2226.15 Billion

HTTP Error 401:

Tesla (TSLA): \$1037.66 Billion

NVIDIA (NVDA): \$3508.16 Billion

Netflix (NFLX): \$524.05 Billion

Meta (META): \$1481.48 Billion

Microsoft (MSFT): \$3548.29 Billion

```
In [26]: techStocks = {
    'AAPL': 'Apple',
    'PLTR': 'Palantir',
    'GOOGL': 'Google',
    'AMZN': 'Amazon',
    'TSLA': 'Tesla',
    'NVDA': 'NVIDIA',
    'NFLX': 'Netflix',
    'META': 'Meta',
    'MSFT': 'Microsoft'
}

end_date = datetime.now()
start_date = end_date - timedelta(days=3650)

stockData = {}
for symbol, name in techStocks.items():
```

```

    ticker = yf.Ticker(symbol)
    data = ticker.history(start=start_date, end=end_date)
    shareCount = ticker.info.get('sharesOutstanding')
    marketVal = data['Close'] * shareCount
    stockData[name] = marketVal

plt.figure(figsize=(15,10))
for name, prices in stockData.items():
    if len(prices) > 0:
        plt.plot(prices.index, prices.values / 1e9, linewidth=2, label=name)

plt.title('Market Cap of Major Tech Companies (10 years)', fontsize=16, fontweight=
plt.xlabel("Date", fontsize=12)
plt.ylabel("Market Cap (Billion USD)", fontsize=12)
plt.legend(bbox_to_anchor=(1.05, 1), loc='upper left')
plt.grid(True, alpha=0.3)
plt.xticks(rotation=45)
plt.tight_layout()
plt.show()

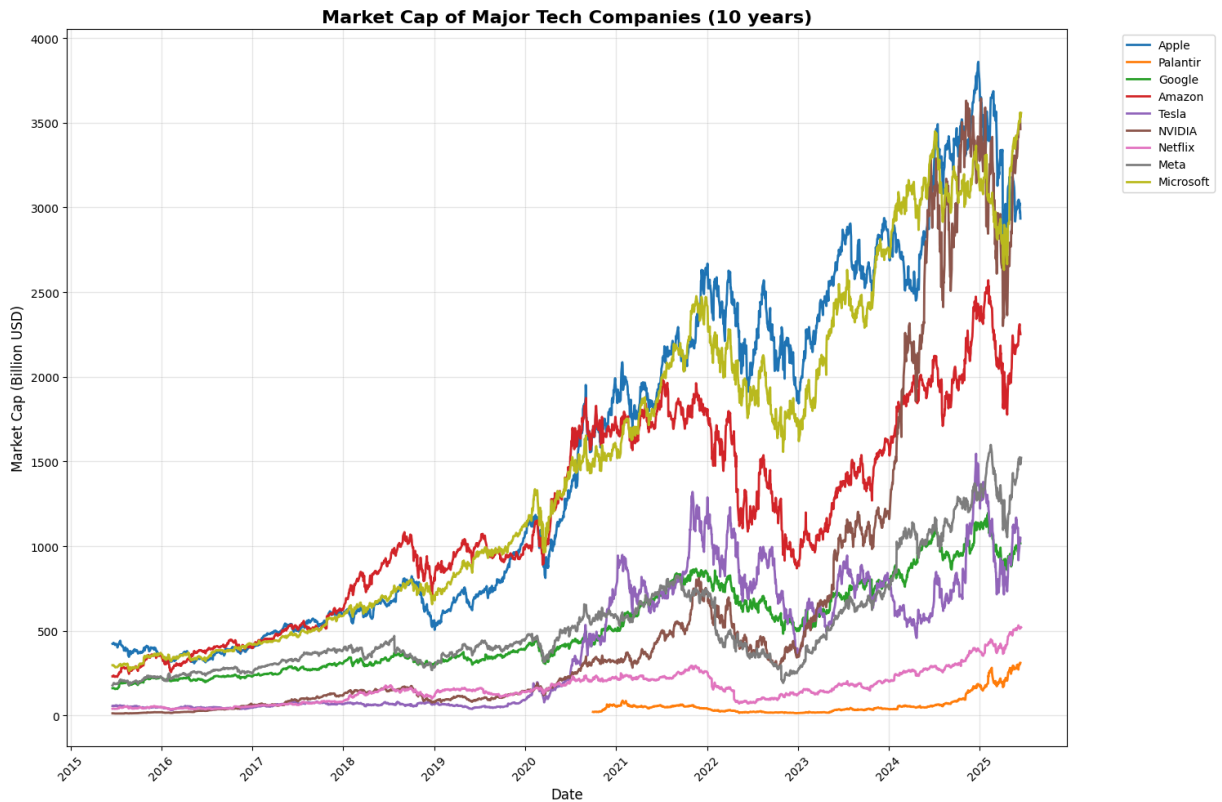
print("\nEstimated Current Market Capitalizations:")
print("-" * 50)
for symbol, name in techStocks.items():
    ticker = yf.Ticker(symbol)
    currentPrice = ticker.history(period="1d")['Close'].iloc[-1]
    shares = ticker.info.get('sharesOutstanding')
    marketCap = currentPrice * shares
    print(f"{name:12} ({symbol}): ${marketCap / 1e9:.2f} Billion")

```

```

HTTP Error 401:
HTTP Error 401:
HTTP Error 401:

```



Estimated Current Market Capitalizations:

HTTP Error 401:

Apple (AAPL): \$2963.79 Billion

HTTP Error 401:

Palantir (PLTR): \$321.05 Billion

HTTP Error 401:

Google (GOOGL): \$1027.67 Billion

Amazon (AMZN): \$2300.04 Billion

Tesla (TSLA): \$1060.07 Billion

HTTP Error 401:

NVIDIA (NVDA): \$3529.13 Billion

HTTP Error 401:

Netflix (NFLX): \$520.85 Billion

Meta (META): \$1521.37 Billion

Microsoft (MSFT): \$3557.40 Billion

```
In [13]: techStocks = {
    'AAPL': 'Apple',
    'PLTR': 'Palantir',
    'GOOGL': 'Google',
    'AMZN': 'Amazon',
    'TSLA': 'Tesla',
    'NVDA': 'NVIDIA',
    'NFLX': 'Netflix',
    'META': 'Meta',
    'MSFT': 'Microsoft'
}

end_date = datetime.now()
start_date = end_date - timedelta(days=3650)
```

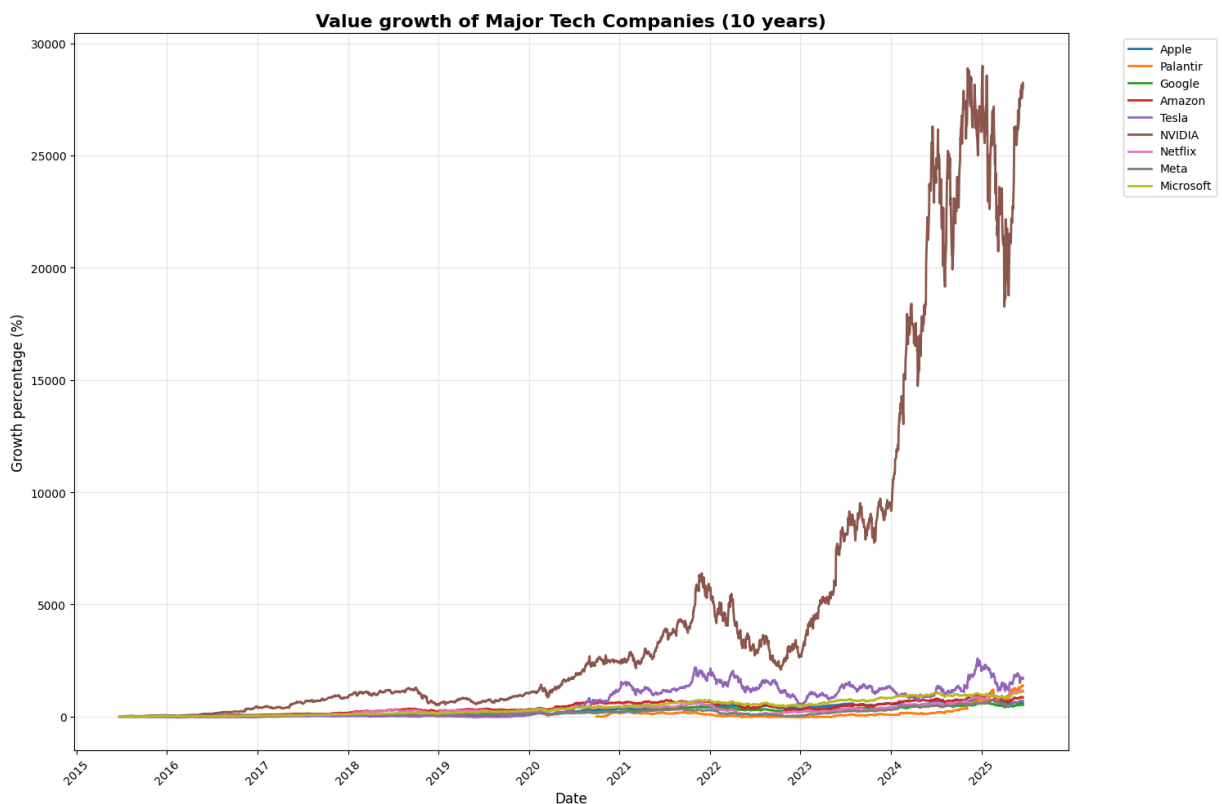
```

stockData = {}
for symbol, name in techStocks.items():
    ticker = yf.Ticker(symbol)
    data = ticker.history(start=start_date, end=end_date)
    stockData[name] = data['Close']

plt.figure(figsize=(15,10))
for name, prices in stockData.items():
    if len(prices) > 0:
        norm = (prices / prices.iloc[0] - 1) * 100
        plt.plot(norm.index, norm.values, linewidth=2, label=name)

plt.title('Value growth of Major Tech Companies (10 years)', fontsize=16, fontweight='bold')
plt.xlabel("Date", fontsize=12)
plt.ylabel("Growth percentage (%)", fontsize=12)
plt.legend(bbox_to_anchor=(1.05, 1), loc='upper left')
plt.grid(True, alpha=0.3)
plt.xticks(rotation=45)
plt.tight_layout()
plt.show()

```



```

In [14]: techStocks = {
    'AAPL': 'Apple',
    'PLTR': 'Palantir',
    'GOOGL': 'Google',
    'AMZN': 'Amazon',
    'TSLA': 'Tesla',
    'NFLX': 'Netflix',
    'META': 'Meta',

```

```

    'MSFT': 'Microsoft'
}

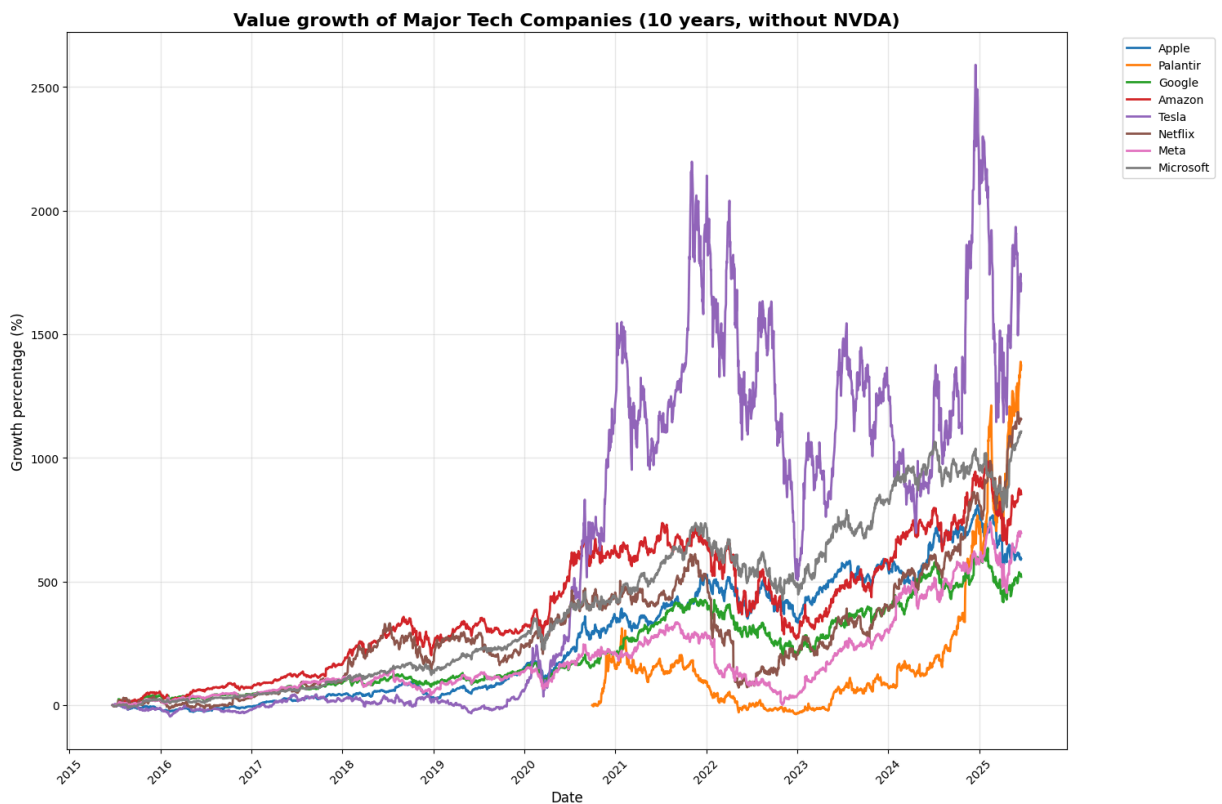
end_date = datetime.now()
start_date = end_date - timedelta(days=3650)

stockData = {}
for symbol, name in techStocks.items():
    ticker = yf.Ticker(symbol)
    data = ticker.history(start=start_date, end=end_date)
    stockData[name] = data['Close']

plt.figure(figsize=(15,10))
for name, prices in stockData.items():
    if len(prices) > 0:
        norm = (prices / prices.iloc[0] - 1) * 100
        plt.plot(norm.index, norm.values, linewidth=2, label=name)

plt.title('Value growth of Major Tech Companies (10 years, without NVDA)', fontsize=14)
plt.xlabel("Date", fontsize=12)
plt.ylabel("Growth percentage (%)", fontsize=12)
plt.legend(bbox_to_anchor=(1.05, 1), loc='upper left')
plt.grid(True, alpha=0.3)
plt.xticks(rotation=45)
plt.tight_layout()
plt.show()

```



```

In [15]: techStocks = {
    'AAPL': 'Apple',
    'PLTR': 'Palantir',

```

```

    'GOOGL': 'Google',
    'AMZN': 'Amazon',
    'NVDA': 'Nvidia',
    'TSLA': 'Tesla',
    'NFLX': 'Netflix',
    'META': 'Meta',
    'MSFT': 'Microsoft'
}

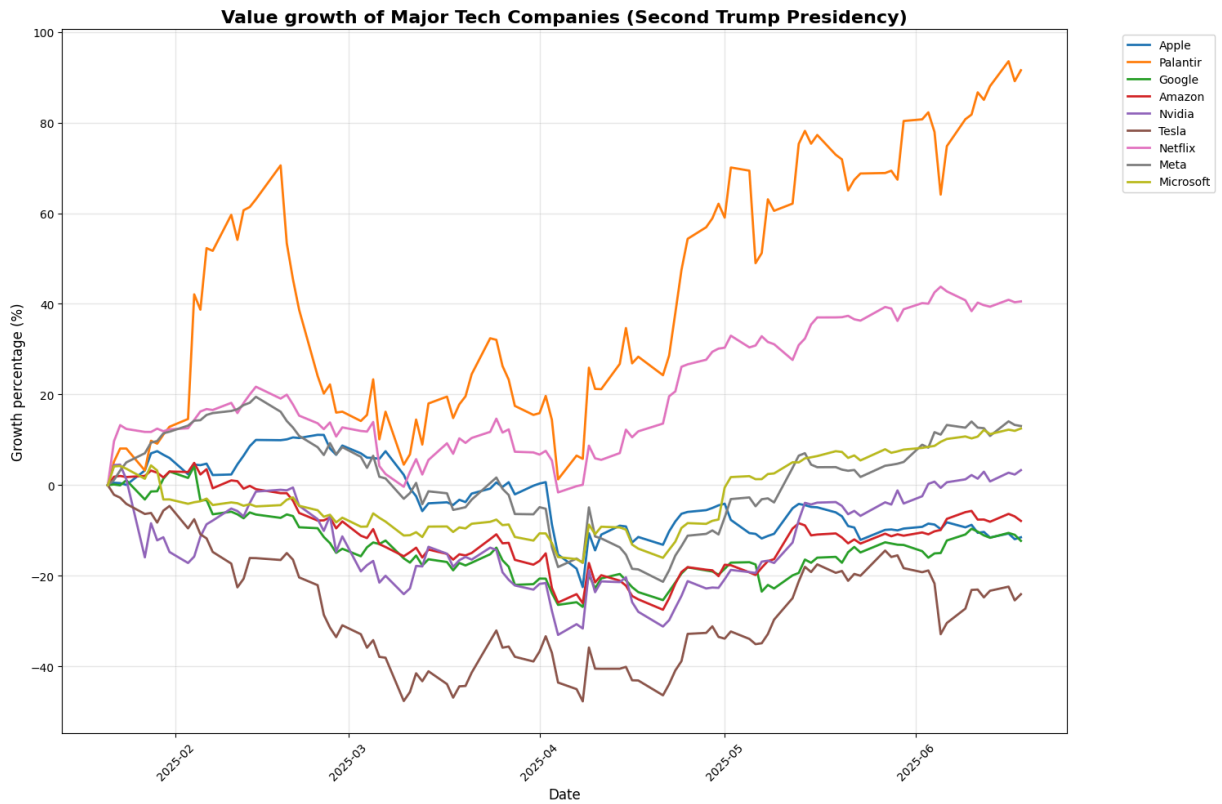
end_date = datetime.now()
start_date = datetime(2025, 1, 20)

stockData = {}
for symbol, name in techStocks.items():
    ticker = yf.Ticker(symbol)
    data = ticker.history(start=start_date, end=end_date)
    stockData[name] = data['Close']

plt.figure(figsize=(15,10))
for name, prices in stockData.items():
    if len(prices) > 0:
        norm = (prices / prices.iloc[0] - 1) * 100
        plt.plot(norm.index, norm.values, linewidth=2, label=name)

plt.title('Value growth of Major Tech Companies (Second Trump Presidency)', fontsize=12)
plt.xlabel("Date", fontsize=12)
plt.ylabel("Growth percentage (%)", fontsize=12)
plt.legend(bbox_to_anchor=(1.05, 1), loc='upper left')
plt.grid(True, alpha=0.3)
plt.xticks(rotation=45)
plt.tight_layout()
plt.show()

```



```
In [16]: techStocks = {
    'AAPL': 'Apple',
    'PLTR': 'Palantir',
    'GOOGL': 'Google',
    'AMZN': 'Amazon',
    'NVDA': 'Nvidia',
    'TSLA': 'Tesla',
    'NFLX': 'Netflix',
    'META': 'Meta',
    'MSFT': 'Microsoft'
}

end_date = datetime(2025, 1, 20)
start_date = datetime(2021, 1, 20)

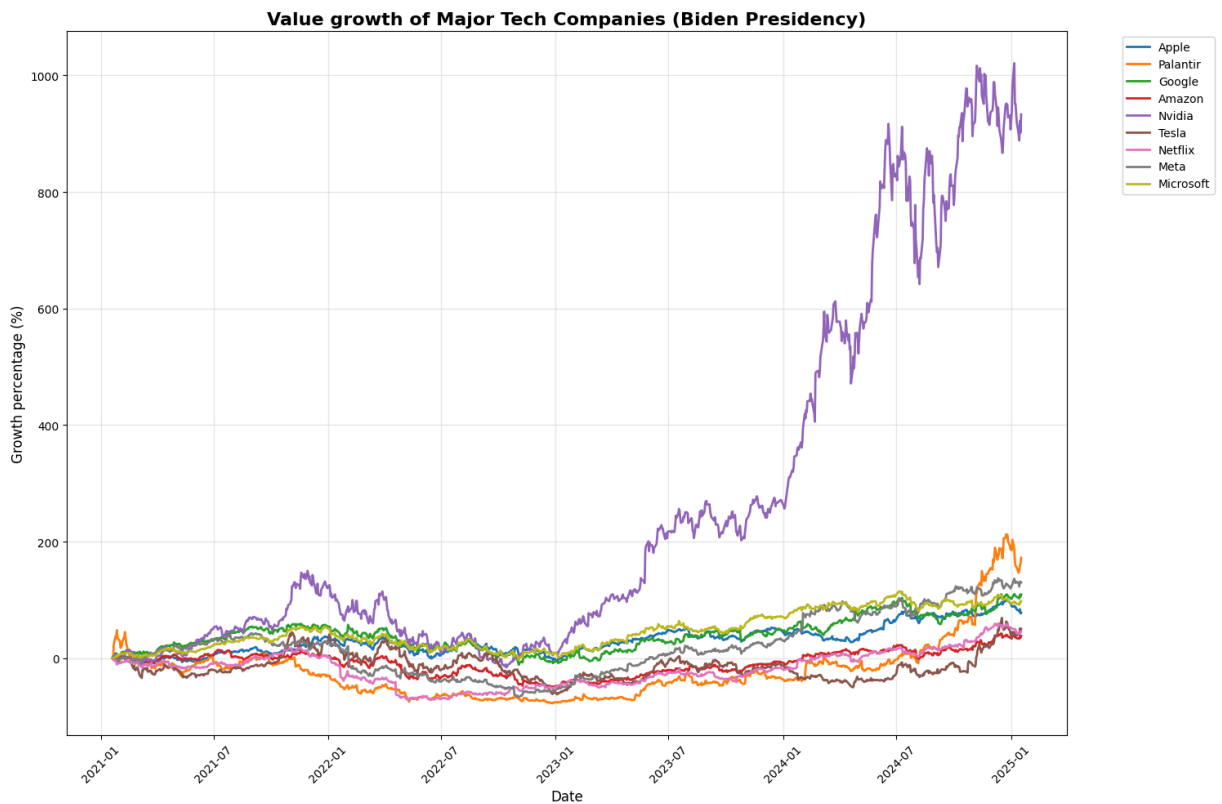
stockData = {}
for symbol, name in techStocks.items():
    ticker = yf.Ticker(symbol)
    data = ticker.history(start=start_date, end=end_date)
    stockData[name] = data['Close']

plt.figure(figsize=(15,10))
for name, prices in stockData.items():
    if len(prices) > 0:
        norm = (prices / prices.iloc[0] - 1) * 100
        plt.plot(norm.index, norm.values, linewidth=2, label=name)

plt.title('Value growth of Major Tech Companies (Biden Presidency)', fontsize=16, f
plt.xlabel("Date", fontsize=12)
plt.ylabel("Growth percentage (%)", fontsize=12)
```



```
plt.legend(bbox_to_anchor=(1.05, 1), loc='upper left')
plt.grid(True, alpha=0.3)
plt.xticks(rotation=45)
plt.tight_layout()
plt.show()
```



```
In [17]: techStocks = {
    'AAPL': 'Apple',
    'PLTR': 'Palantir',
    'GOOGL': 'Google',
    'AMZN': 'Amazon',
    'TSLA': 'Tesla',
    'NFLX': 'Netflix',
    'META': 'Meta',
    'MSFT': 'Microsoft'
}

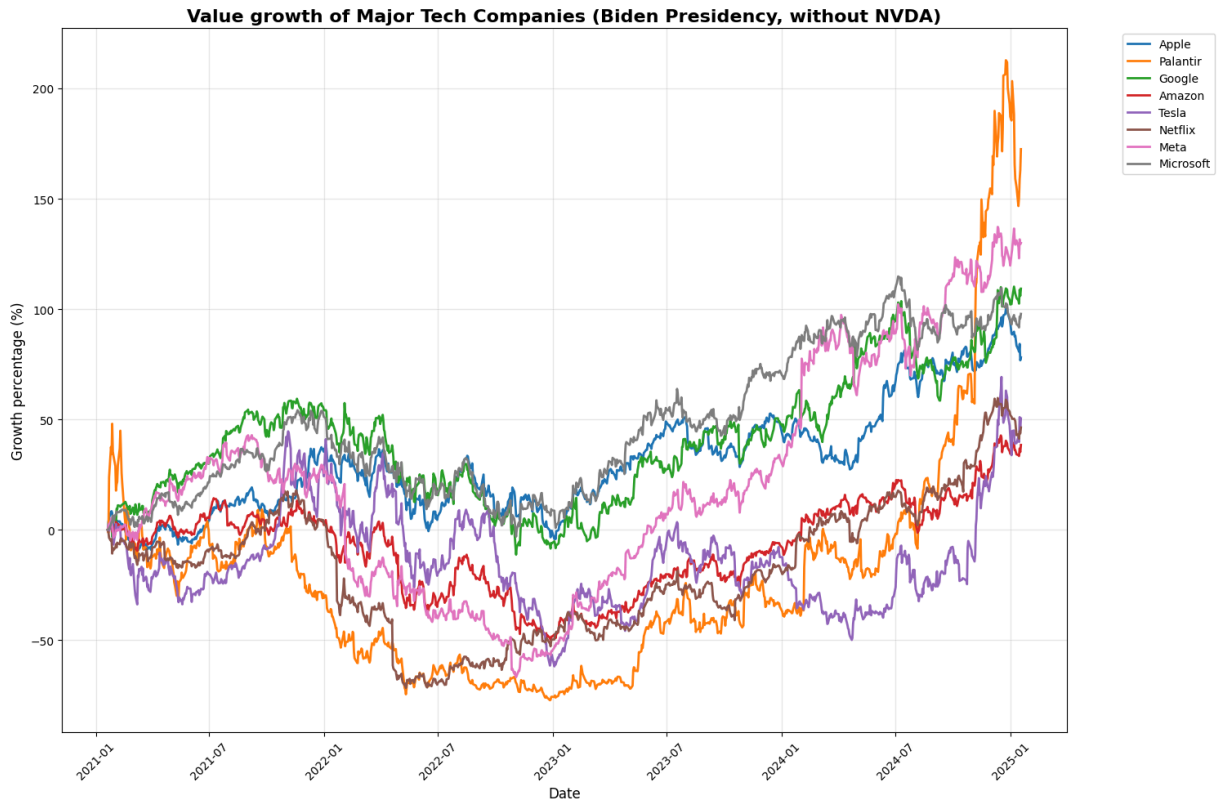
end_date = datetime(2025, 1, 20)
start_date = datetime(2021, 1, 20)

stockData = {}
for symbol, name in techStocks.items():
    ticker = yf.Ticker(symbol)
    data = ticker.history(start=start_date, end=end_date)
    stockData[name] = data['Close']

plt.figure(figsize=(15,10))
for name, prices in stockData.items():
    if len(prices) > 0:
        norm = (prices / prices.iloc[0] - 1) * 100
```

```
plt.plot(norm.index, norm.values, linewidth=2, label=name)

plt.title('Value growth of Major Tech Companies (Biden Presidency, without NVDA)',
plt.xlabel("Date", fontsize=12)
plt.ylabel("Growth percentage (%)", fontsize=12)
plt.legend(bbox_to_anchor=(1.05, 1), loc='upper left')
plt.grid(True, alpha=0.3)
plt.xticks(rotation=45)
plt.tight_layout()
plt.show()
```



```
In [18]: techStocks = {
    'AAPL': 'Apple',
    'PLTR': 'Palantir',
    'GOOGL': 'Google',
    'AMZN': 'Amazon',
    'TSLA': 'Tesla',
    'NVDA': 'Nvidia',
    'NFLX': 'Netflix',
    'META': 'Meta',
    'MSFT': 'Microsoft'
}

end_date = datetime(2021, 1, 20)
start_date = datetime(2017, 1, 20)

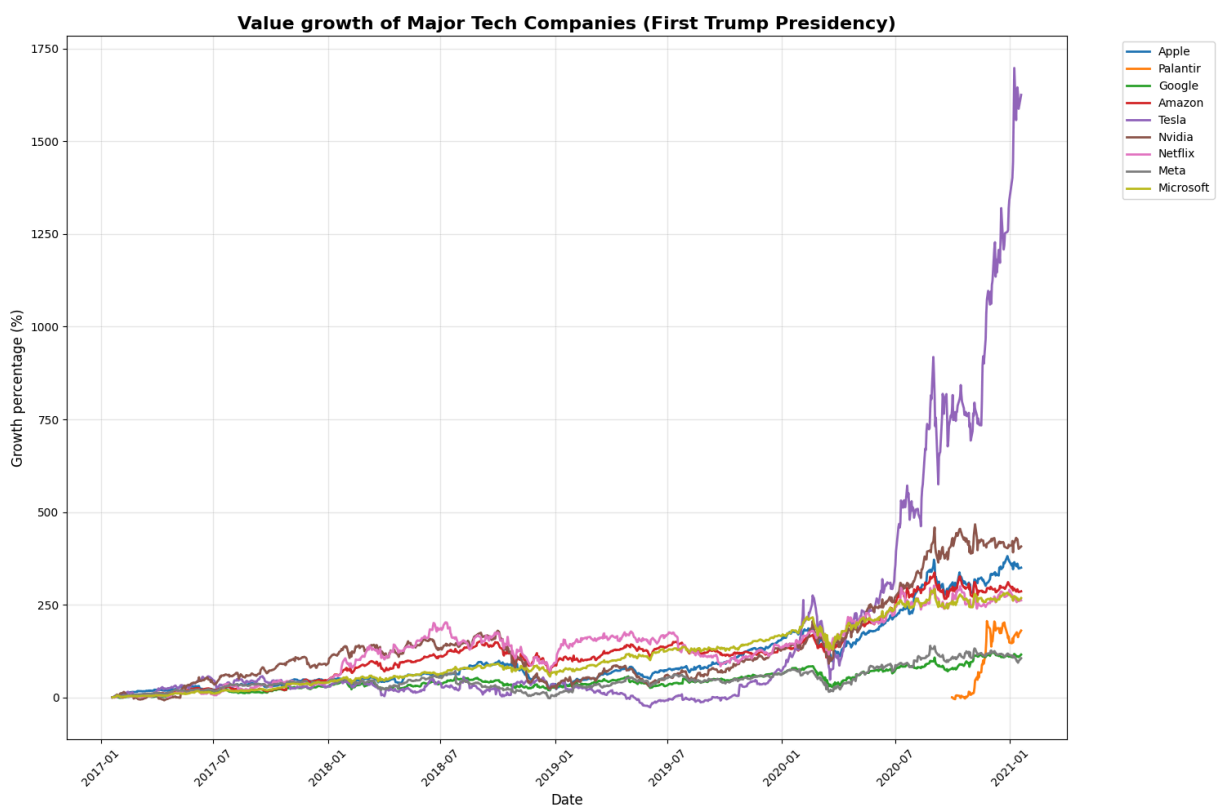
stockData = {}
for symbol, name in techStocks.items():
    ticker = yf.Ticker(symbol)
    data = ticker.history(start=start_date, end=end_date)
    stockData[name] = data['Close']
```

```

plt.figure(figsize=(15,10))
for name, prices in stockData.items():
    if len(prices) > 0:
        norm = (prices / prices.iloc[0] - 1) * 100
        plt.plot(norm.index, norm.values, linewidth=2, label=name)

plt.title('Value growth of Major Tech Companies (First Trump Presidency)', fontsize=14)
plt.xlabel("Date", fontsize=12)
plt.ylabel("Growth percentage (%)", fontsize=12)
plt.legend(bbox_to_anchor=(1.05, 1), loc='upper left')
plt.grid(True, alpha=0.3)
plt.xticks(rotation=45)
plt.tight_layout()
plt.show()

```



```

In [19]: techStocks = {
    'AAPL': 'Apple',
    'PLTR': 'Palantir',
    'GOOGL': 'Google',
    'AMZN': 'Amazon',
    'NVDA': 'Nvidia',
    'NFLX': 'Netflix',
    'META': 'Meta',
    'MSFT': 'Microsoft'
}

end_date = datetime(2021, 1, 20)
start_date = datetime(2017, 1, 20)

```

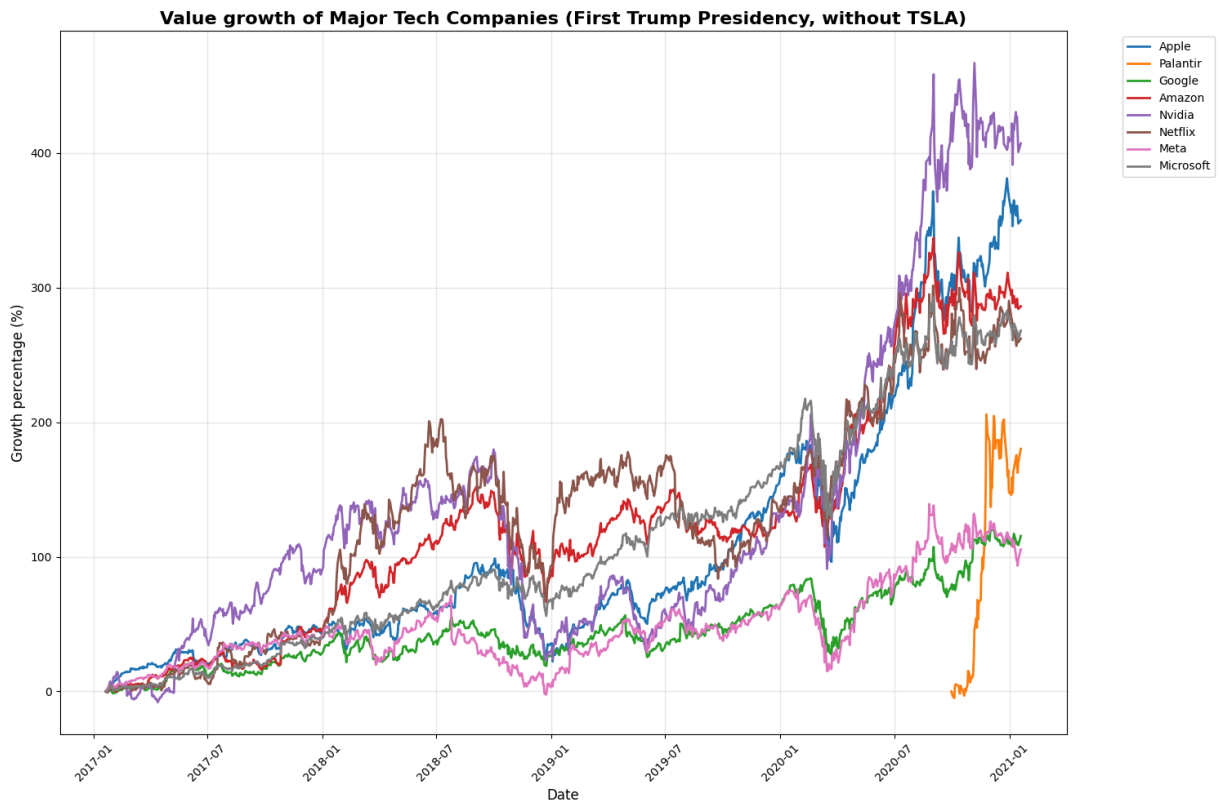
```

stockData = {}
for symbol, name in techStocks.items():
    ticker = yf.Ticker(symbol)
    data = ticker.history(start=start_date, end=end_date)
    stockData[name] = data['Close']

plt.figure(figsize=(15,10))
for name, prices in stockData.items():
    if len(prices) > 0:
        norm = (prices / prices.iloc[0] - 1) * 100
        plt.plot(norm.index, norm.values, linewidth=2, label=name)

plt.title('Value growth of Major Tech Companies (First Trump Presidency, without TS
plt.xlabel("Date", fontsize=12)
plt.ylabel("Growth percentage (%)", fontsize=12)
plt.legend(bbox_to_anchor=(1.05, 1), loc='upper left')
plt.grid(True, alpha=0.3)
plt.xticks(rotation=45)
plt.tight_layout()
plt.show()

```



```

In [20]: techStocks = {
    'AAPL': 'Apple',
    'GOOGL': 'Google',
    'AMZN': 'Amazon',
    'TSLA': 'Tesla',
    'NVDA': 'Nvidia',
    'NFLX': 'Netflix',
    'META': 'Meta',
    'MSFT': 'Microsoft',

```

```

}

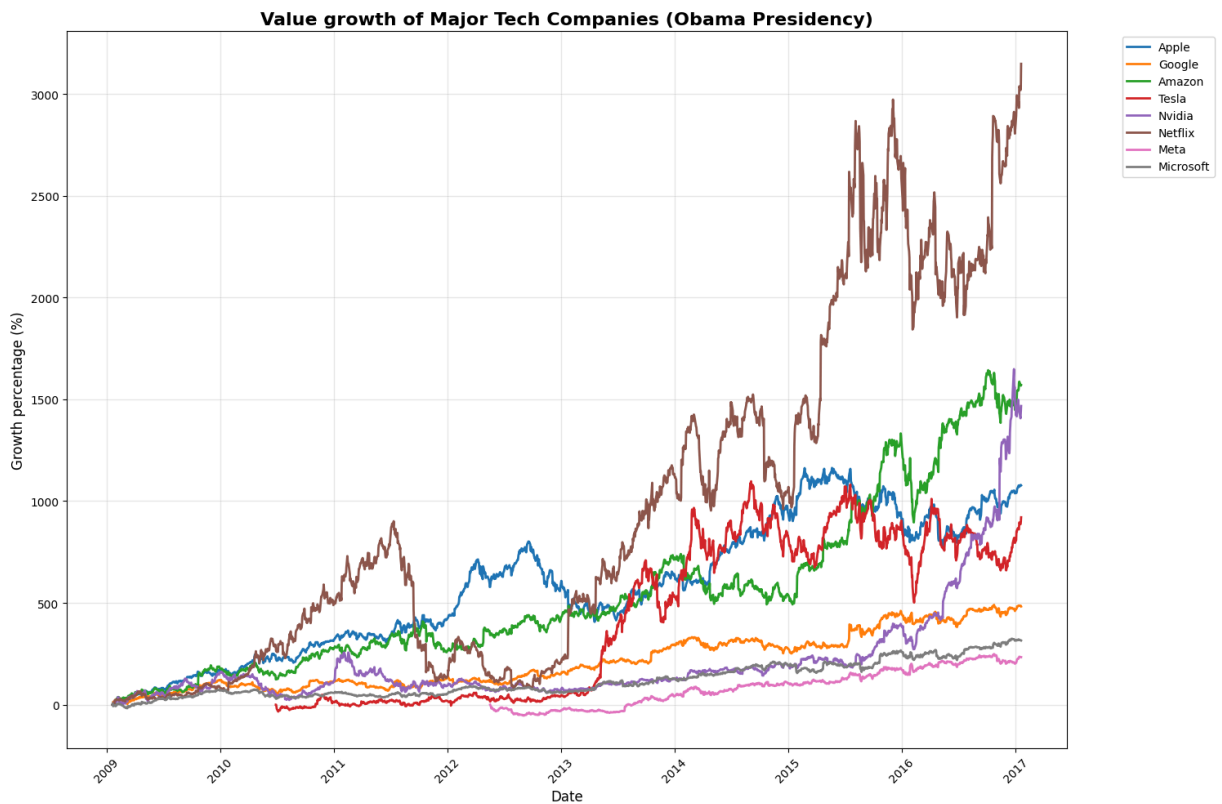
end_date = datetime(2017, 1, 20)
start_date = datetime(2009, 1, 20)

stockData = {}
for symbol, name in techStocks.items():
    ticker = yf.Ticker(symbol)
    data = ticker.history(start=start_date, end=end_date)
    stockData[name] = data['Close']

plt.figure(figsize=(15,10))
for name, prices in stockData.items():
    if len(prices) > 0:
        norm = (prices / prices.iloc[0] - 1) * 100
        plt.plot(norm.index, norm.values, linewidth=2, label=name)

plt.title('Value growth of Major Tech Companies (Obama Presidency)', fontsize=16, f
plt.xlabel("Date", fontsize=12)
plt.ylabel("Growth percentage (%)", fontsize=12)
plt.legend(bbox_to_anchor=(1.05, 1), loc='upper left')
plt.grid(True, alpha=0.3)
plt.xticks(rotation=45)
plt.tight_layout()
plt.show()

```



```

In [21]: techStocks = {
    'AAPL': 'Apple',
    'GOOGL': 'Google',
    'AMZN': 'Amazon',

```

```

    'NVDA': 'Nvidia',
    'NFLX': 'Netflix',
    'MSFT': 'Microsoft'
}

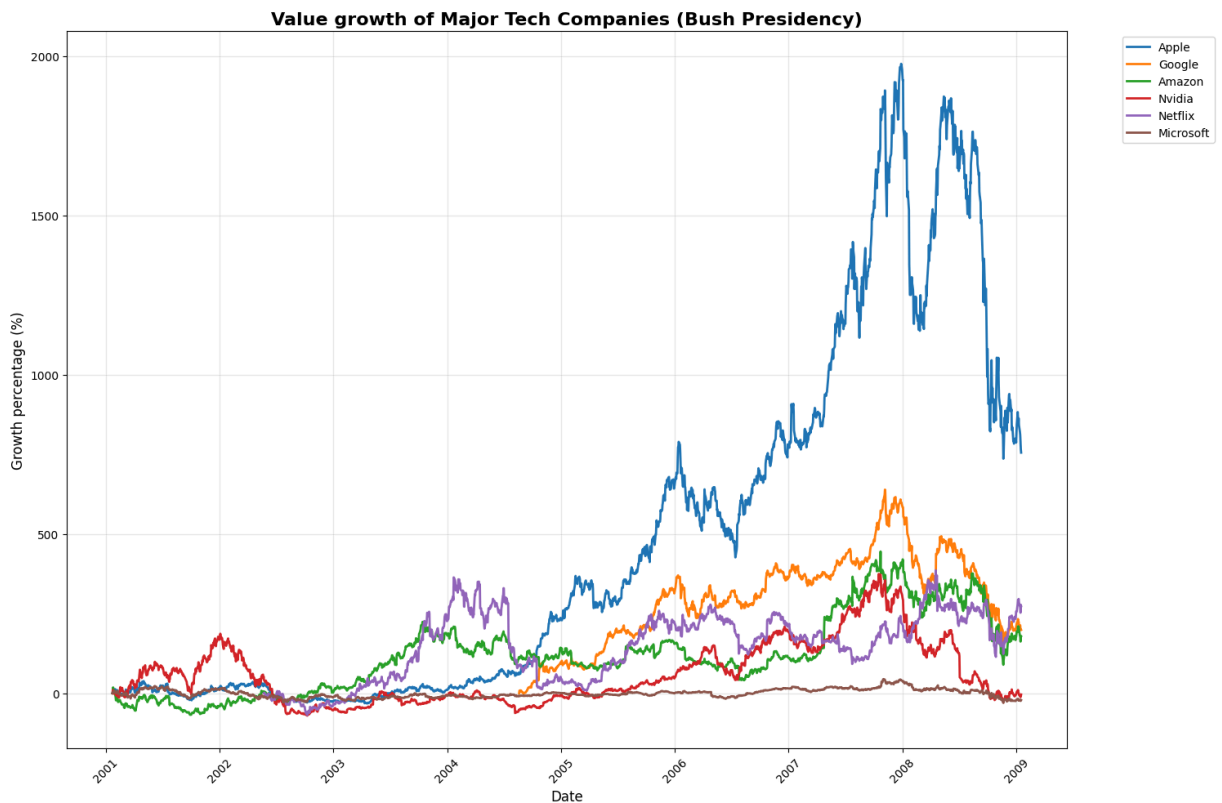
end_date = datetime(2009, 1, 20)
start_date = datetime(2001, 1, 20)

stockData = {}
for symbol, name in techStocks.items():
    ticker = yf.Ticker(symbol)
    data = ticker.history(start=start_date, end=end_date)
    stockData[name] = data['Close']

plt.figure(figsize=(15,10))
for name, prices in stockData.items():
    if len(prices) > 0:
        norm = (prices / prices.iloc[0] - 1) * 100
        plt.plot(norm.index, norm.values, linewidth=2, label=name)

plt.title('Value growth of Major Tech Companies (Bush Presidency)', fontsize=16, fo
plt.xlabel("Date", fontsize=12)
plt.ylabel("Growth percentage (%)", fontsize=12)
plt.legend(bbox_to_anchor=(1.05, 1), loc='upper left')
plt.grid(True, alpha=0.3)
plt.xticks(rotation=45)
plt.tight_layout()
plt.show()

```



```

In [22]: techStocks = {
    'GOOGL': 'Google',
    'AMZN': 'Amazon',
    'NVDA': 'Nvidia',
    'NFLX': 'Netflix',
    'MSFT': 'Microsoft'
}

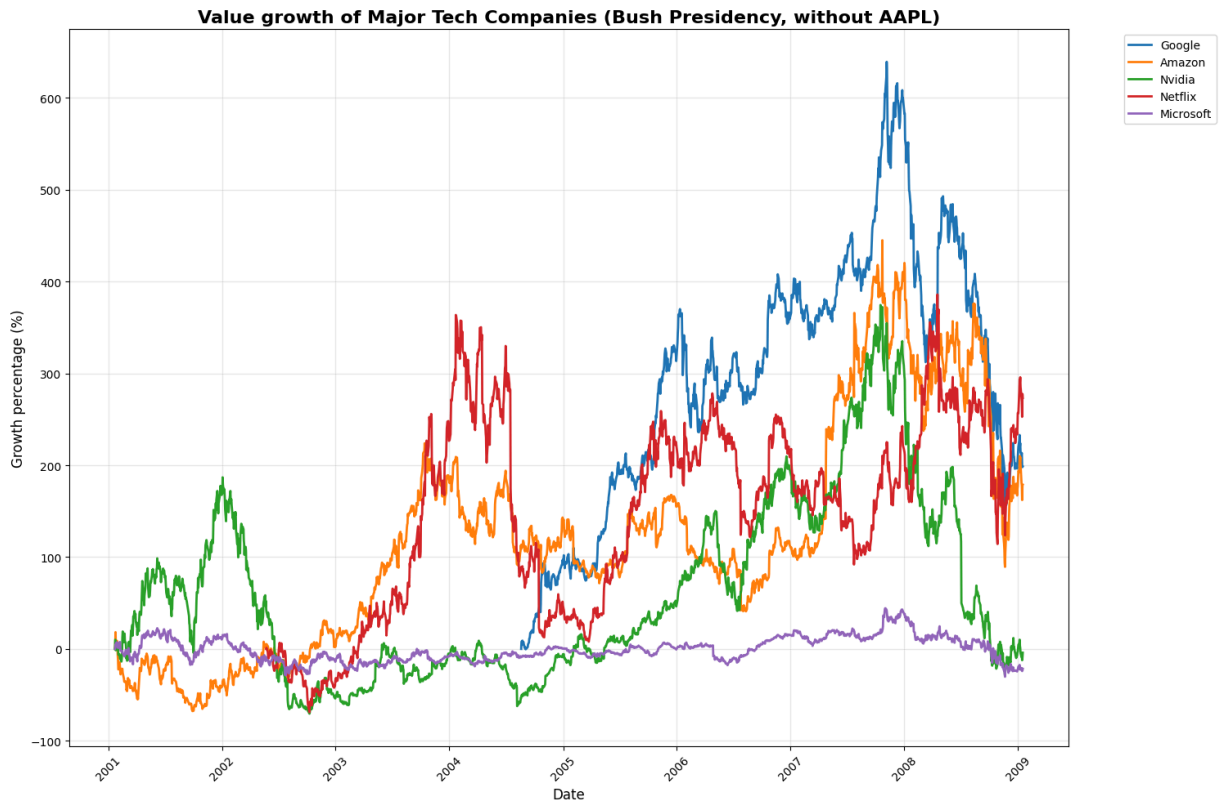
end_date = datetime(2009, 1, 20)
start_date = datetime(2001, 1, 20)

stockData = {}
for symbol, name in techStocks.items():
    ticker = yf.Ticker(symbol)
    data = ticker.history(start=start_date, end=end_date)
    stockData[name] = data['Close']

plt.figure(figsize=(15,10))
for name, prices in stockData.items():
    if len(prices) > 0:
        norm = (prices / prices.iloc[0] - 1) * 100
        plt.plot(norm.index, norm.values, linewidth=2, label=name)

plt.title('Value growth of Major Tech Companies (Bush Presidency, without AAPL)', f
plt.xlabel("Date", fontsize=12)
plt.ylabel("Growth percentage (%)", fontsize=12)
plt.legend(bbox_to_anchor=(1.05, 1), loc='upper left')
plt.grid(True, alpha=0.3)
plt.xticks(rotation=45)
plt.tight_layout()
plt.show()

```



```
In [23]: techStocks = {
    'AAPL': 'Apple',
    'AMZN': 'Amazon',
    'MSFT': 'Microsoft',
    'IBM': 'IBM',
    'CSCO': 'Cisco',
    'INTC': 'Intel'
}

end_date = datetime(2001, 1, 20)
start_date = datetime(1993, 1, 20)

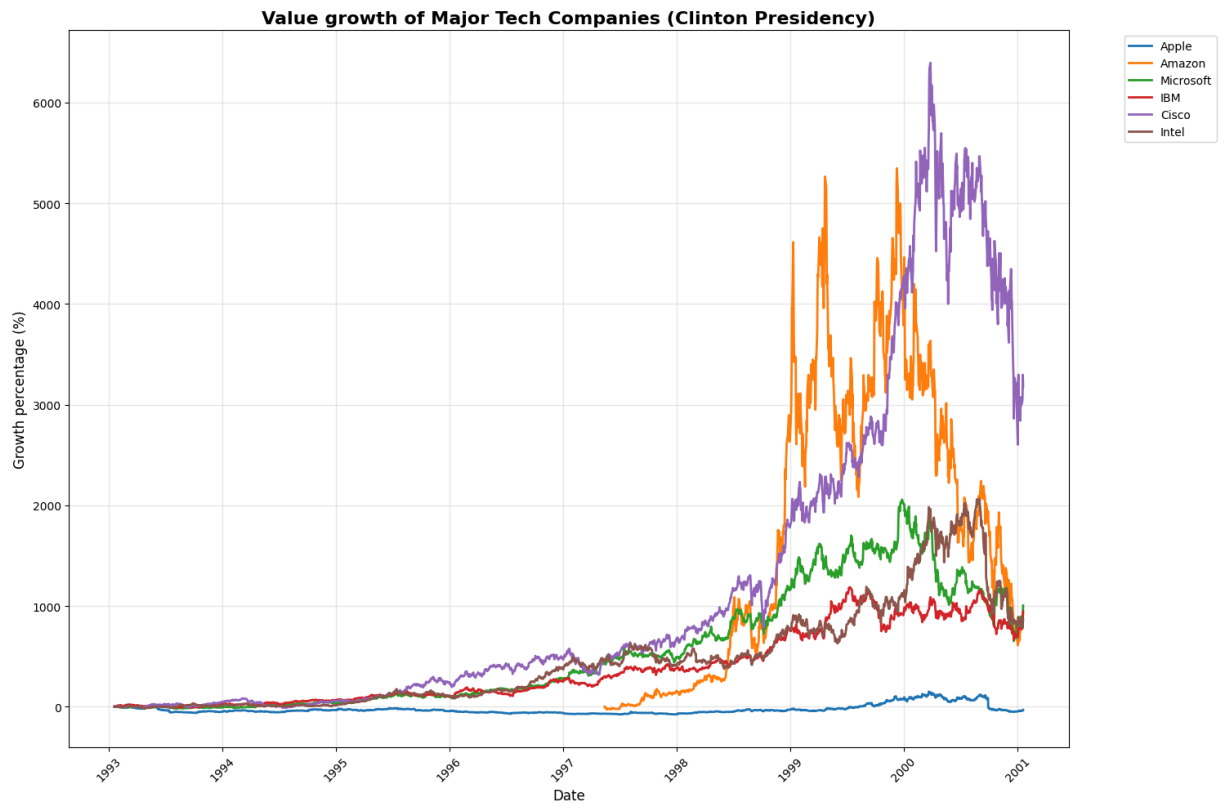
stockData = {}
for symbol, name in techStocks.items():
    ticker = yf.Ticker(symbol)
    data = ticker.history(start=start_date, end=end_date)
    stockData[name] = data['Close']

plt.figure(figsize=(15,10))
for name, prices in stockData.items():
    if len(prices) > 0:
        norm = (prices / prices.iloc[0] - 1) * 100
        plt.plot(norm.index, norm.values, linewidth=2, label=name)

plt.title('Value growth of Major Tech Companies (Clinton Presidency)', fontsize=16,
plt.xlabel("Date", fontsize=12)
plt.ylabel("Growth percentage (%)", fontsize=12)
plt.legend(bbox_to_anchor=(1.05, 1), loc='upper left')
plt.grid(True, alpha=0.3)
plt.xticks(rotation=45)
```



```
plt.tight_layout()
plt.show()
```



```
In [14]: techStocks = {
    'AAPL': 'Apple',
    'PLTR': 'Palantir',
    'GOOGL': 'Google',
    'AMZN': 'Amazon',
    'TSLA': 'Tesla',
    'NVDA': 'NVIDIA',
    'NFLX': 'Netflix',
    'META': 'Meta',
    'MSFT': 'Microsoft'
}

end_date = datetime.now()
start_date = datetime(2025, 1, 20)

stockData = {}
for symbol, name in techStocks.items():
    ticker = yf.Ticker(symbol)
    data = ticker.history(start=start_date, end=end_date)
    shareCount = ticker.info.get('sharesOutstanding')
    marketVal = data['Close'] * shareCount
    stockData[name] = marketVal

plt.figure(figsize=(15,10))
for name, prices in stockData.items():
    if len(prices) > 0:
        plt.plot(prices.index, prices.values / 1e9, linewidth=2, label=name)
```

```

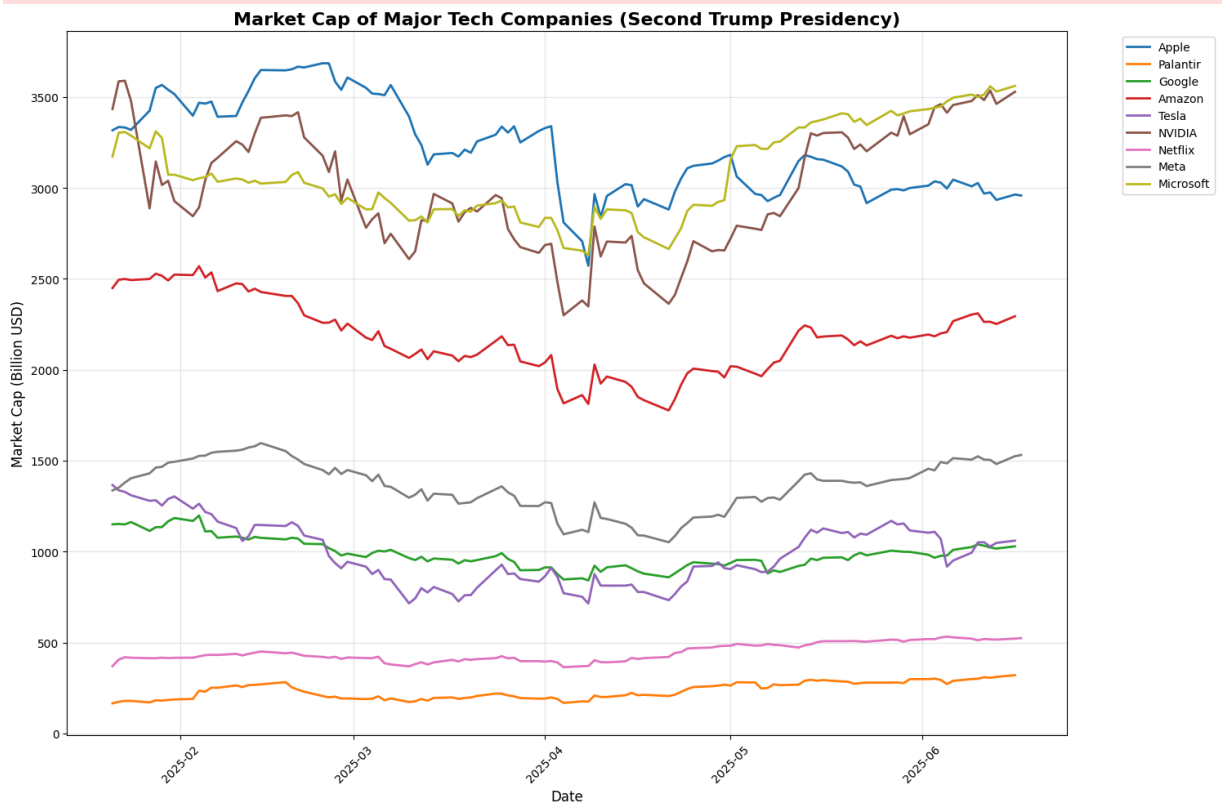
plt.title('Market Cap of Major Tech Companies (Second Trump Presidency)', fontsize=
plt.xlabel("Date", fontsize=12)
plt.ylabel("Market Cap (Billion USD)", fontsize=12)
plt.legend(bbox_to_anchor=(1.05, 1), loc='upper left')
plt.grid(True, alpha=0.3)
plt.xticks(rotation=45)
plt.tight_layout()
plt.show()

print("\nEstimated Current Market Capitalizations:")
print("-" * 50)
for symbol, name in techStocks.items():
    ticker = yf.Ticker(symbol)
    currentPrice = ticker.history(period="1d")['Close'].iloc[-1]
    shares = ticker.info.get('sharesOutstanding')
    marketCap = currentPrice * shares
    print(f"{name:12} ({symbol}): ${marketCap / 1e9:.2f} Billion")

```

HTTP Error 401:

HTTP Error 401:



Estimated Current Market Capitalizations:

```

-----
Apple      (AAPL): $2959.08 Billion
Palantir   (PLTR): $316.35 Billion
Google     (GOOGL): $1031.07 Billion
Amazon     (AMZN): $2304.27 Billion
Tesla      (TSLA): $1028.88 Billion
NVIDIA     (NVDA): $3529.98 Billion
Netflix    (NFLX): $523.84 Billion
Meta       (META): $1531.10 Billion
Microsoft  (MSFT): $3550.15 Billion

```

```

In [15]: techStocks = {
    'AAPL': 'Apple',
    'PLTR': 'Palantir',
    'GOOGL': 'Google',
    'AMZN': 'Amazon',
    'TSLA': 'Tesla',
    'NVDA': 'NVIDIA',
    'NFLX': 'Netflix',
    'META': 'Meta',
    'MSFT': 'Microsoft'
}

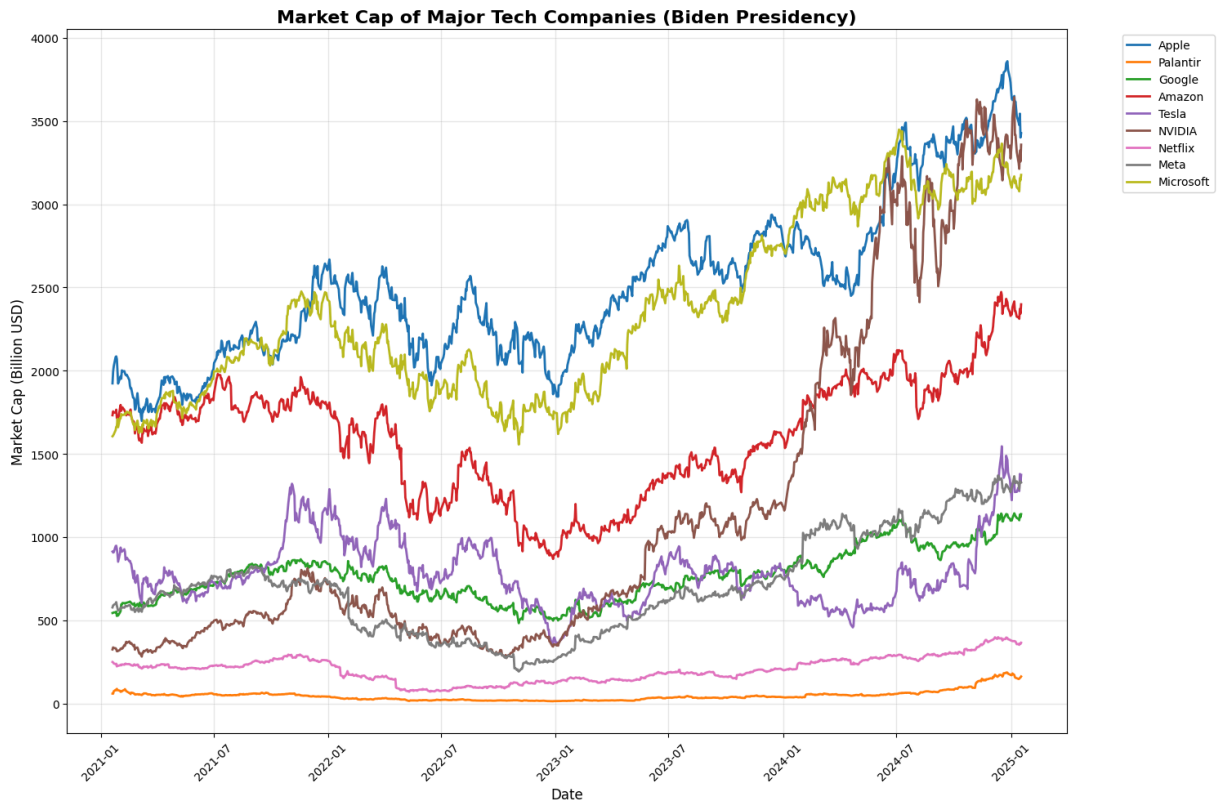
end_date = datetime(2025, 1, 20)
start_date = datetime(2021, 1, 20)

stockData = {}
for symbol, name in techStocks.items():
    ticker = yf.Ticker(symbol)
    data = ticker.history(start=start_date, end=end_date)
    shareCount = ticker.info.get('sharesOutstanding')
    marketVal = data['Close'] * shareCount
    stockData[name] = marketVal

plt.figure(figsize=(15,10))
for name, prices in stockData.items():
    if len(prices) > 0:
        plt.plot(prices.index, prices.values / 1e9, linewidth=2, label=name)

plt.title('Market Cap of Major Tech Companies (Biden Presidency)', fontsize=16, font
plt.xlabel("Date", fontsize=12)
plt.ylabel("Market Cap (Billion USD)", fontsize=12)
plt.legend(bbox_to_anchor=(1.05, 1), loc='upper left')
plt.grid(True, alpha=0.3)
plt.xticks(rotation=45)
plt.tight_layout()
plt.show()

```



```
In [16]: techStocks = {
    'AAPL': 'Apple',
    'PLTR': 'Palantir',
    'GOOGL': 'Google',
    'AMZN': 'Amazon',
    'TSLA': 'Tesla',
    'NVDA': 'NVIDIA',
    'NFLX': 'Netflix',
    'META': 'Meta',
    'MSFT': 'Microsoft'
}

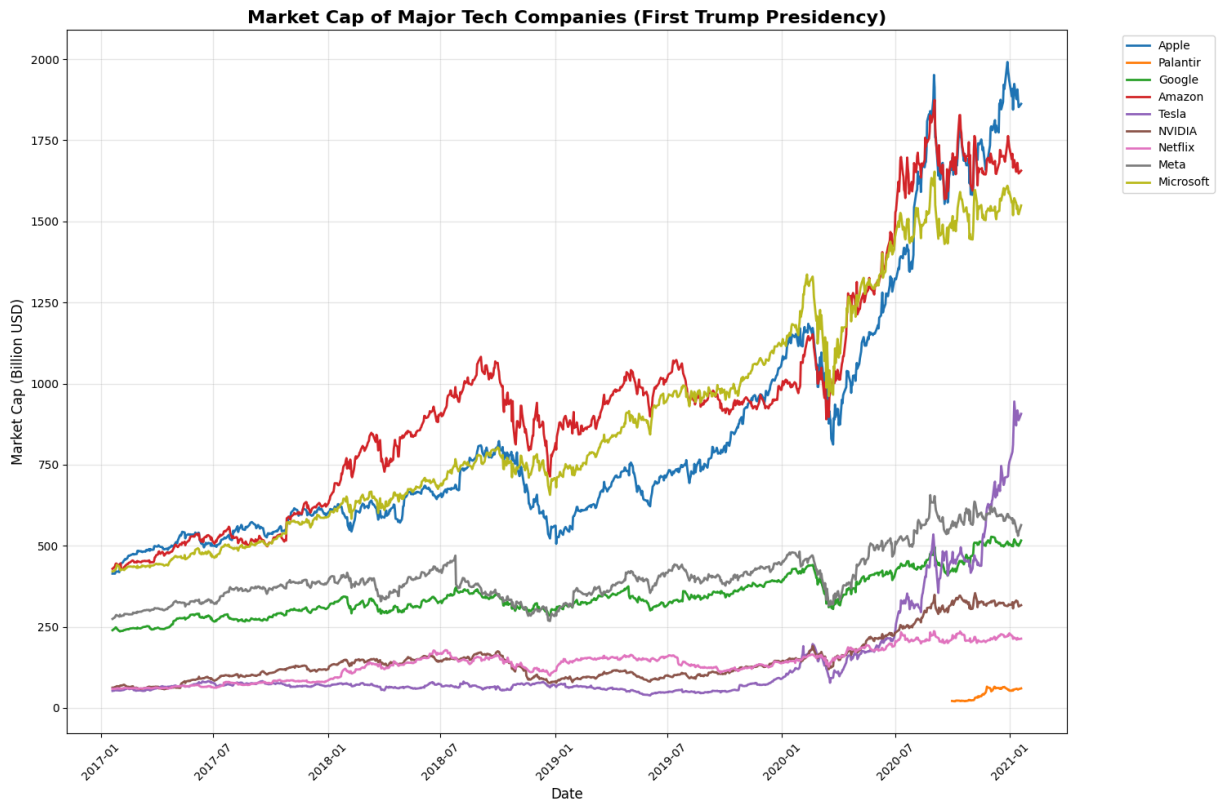
end_date = datetime(2021, 1, 20)
start_date = datetime(2017, 1, 20)

stockData = {}
for symbol, name in techStocks.items():
    ticker = yf.Ticker(symbol)
    data = ticker.history(start=start_date, end=end_date)
    shareCount = ticker.info.get('sharesOutstanding')
    marketVal = data['Close'] * shareCount
    stockData[name] = marketVal

plt.figure(figsize=(15,10))
for name, prices in stockData.items():
    if len(prices) > 0:
        plt.plot(prices.index, prices.values / 1e9, linewidth=2, label=name)

plt.title('Market Cap of Major Tech Companies (First Trump Presidency)', fontsize=12)
plt.xlabel("Date", fontsize=12)
```

```
plt.ylabel("Market Cap (Billion USD)", fontsize=12)
plt.legend(bbox_to_anchor=(1.05, 1), loc='upper left')
plt.grid(True, alpha=0.3)
plt.xticks(rotation=45)
plt.tight_layout()
plt.show()
```



```
In [17]: techStocks = {
    'AAPL': 'Apple',
    'GOOGL': 'Google',
    'AMZN': 'Amazon',
    'TSLA': 'Tesla',
    'NVDA': 'NVIDIA',
    'NFLX': 'Netflix',
    'META': 'Meta',
    'MSFT': 'Microsoft'
}

end_date = datetime(2017, 1, 20)
start_date = datetime(2009, 1, 20)

stockData = {}
for symbol, name in techStocks.items():
    ticker = yf.Ticker(symbol)
    data = ticker.history(start=start_date, end=end_date)
    shareCount = ticker.info.get('sharesOutstanding')
    marketVal = data['Close'] * shareCount
    stockData[name] = marketVal

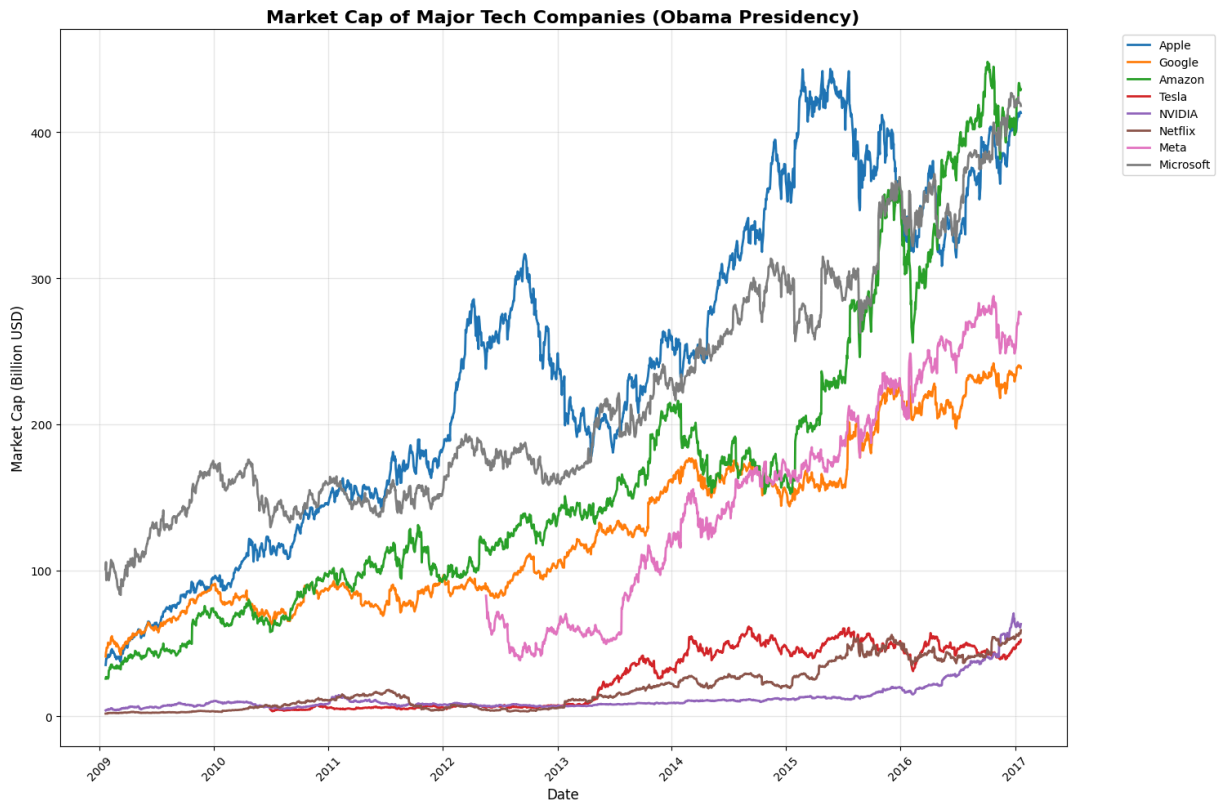
plt.figure(figsize=(15,10))
```

```

for name, prices in stockData.items():
    if len(prices) > 0:
        plt.plot(prices.index, prices.values / 1e9, linewidth=2, label=name)

plt.title('Market Cap of Major Tech Companies (Obama Presidency)', fontsize=16, fontcolor='red')
plt.xlabel("Date", fontsize=12)
plt.ylabel("Market Cap (Billion USD)", fontsize=12)
plt.legend(bbox_to_anchor=(1.05, 1), loc='upper left')
plt.grid(True, alpha=0.3)
plt.xticks(rotation=45)
plt.tight_layout()
plt.show()

```



```

In [24]: techStocks = {
    'AAPL': 'Apple',
    'GOOGL': 'Google',
    'AMZN': 'Amazon',
    'NVDA': 'NVIDIA',
    'NFLX': 'Netflix',
    'MSFT': 'Microsoft',
}

end_date = datetime(2009, 1, 20)
start_date = datetime(2001, 1, 20)

stockData = {}
for symbol, name in techStocks.items():
    ticker = yf.Ticker(symbol)
    data = ticker.history(start=start_date, end=end_date)
    shareCount = ticker.info.get('sharesOutstanding')
    marketVal = data['Close'] * shareCount

```

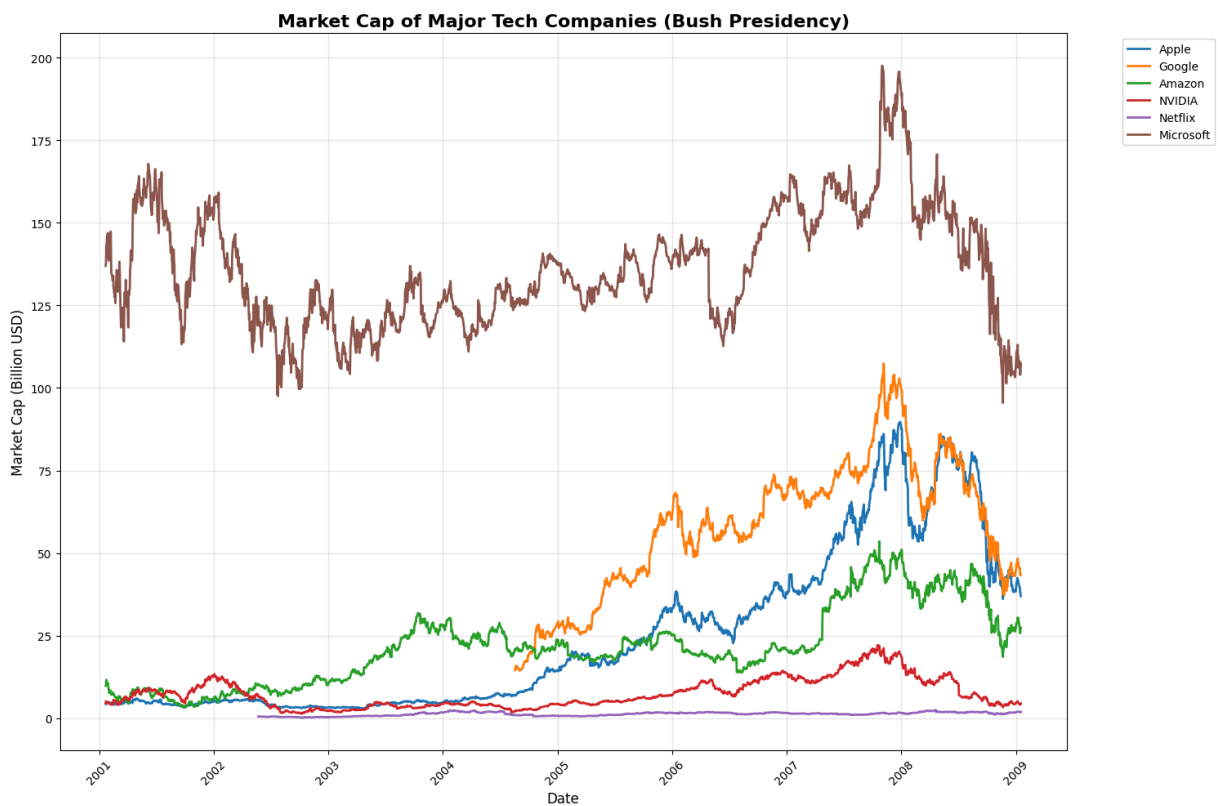
```

stockData[name] = marketVal

plt.figure(figsize=(15,10))
for name, prices in stockData.items():
    if len(prices) > 0:
        plt.plot(prices.index, prices.values / 1e9, linewidth=2, label=name)

plt.title('Market Cap of Major Tech Companies (Bush Presidency)', fontsize=16, font
plt.xlabel("Date", fontsize=12)
plt.ylabel("Market Cap (Billion USD)", fontsize=12)
plt.legend(bbox_to_anchor=(1.05, 1), loc='upper left')
plt.grid(True, alpha=0.3)
plt.xticks(rotation=45)
plt.tight_layout()
plt.show()

```



```

In [22]: techStocks = {
    'AAPL': 'Apple',
    'AMZN': 'Amazon',
    'MSFT': 'Microsoft',
    'IBM': 'IBM',
    'CSCO': 'Cisco',
    'INTC': 'Intel'
}

end_date = datetime(2001, 1, 20)
start_date = datetime(1993, 1, 20)

stockData = {}
for symbol, name in techStocks.items():

```

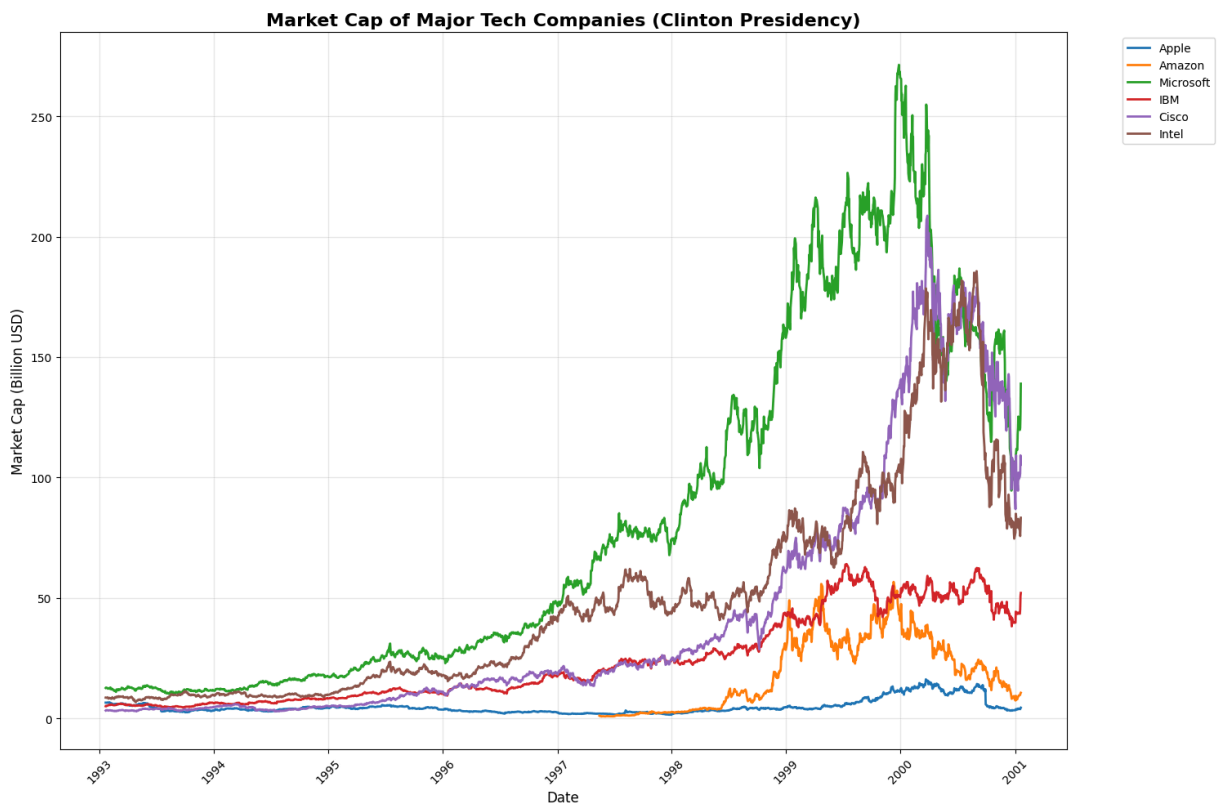
```

ticker = yf.Ticker(symbol)
data = ticker.history(start=start_date, end=end_date)
shareCount = ticker.info.get('sharesOutstanding')
marketVal = data['Close'] * shareCount
stockData[name] = marketVal

plt.figure(figsize=(15,10))
for name, prices in stockData.items():
    if len(prices) > 0:
        plt.plot(prices.index, prices.values / 1e9, linewidth=2, label=name)

plt.title('Market Cap of Major Tech Companies (Clinton Presidency)', fontsize=16, f
plt.xlabel("Date", fontsize=12)
plt.ylabel("Market Cap (Billion USD)", fontsize=12)
plt.legend(bbox_to_anchor=(1.05, 1), loc='upper left')
plt.grid(True, alpha=0.3)
plt.xticks(rotation=45)
plt.tight_layout()
plt.show()

```



```

In [18]: techStocks = {
    'AAPL': 'Apple',
    'GOOGL': 'Google',
    'AMZN': 'Amazon',
    'NFLX': 'Netflix',
    'META': 'Meta',
    'TSLA': 'Tesla'
}

end_date = datetime.now()

```



```

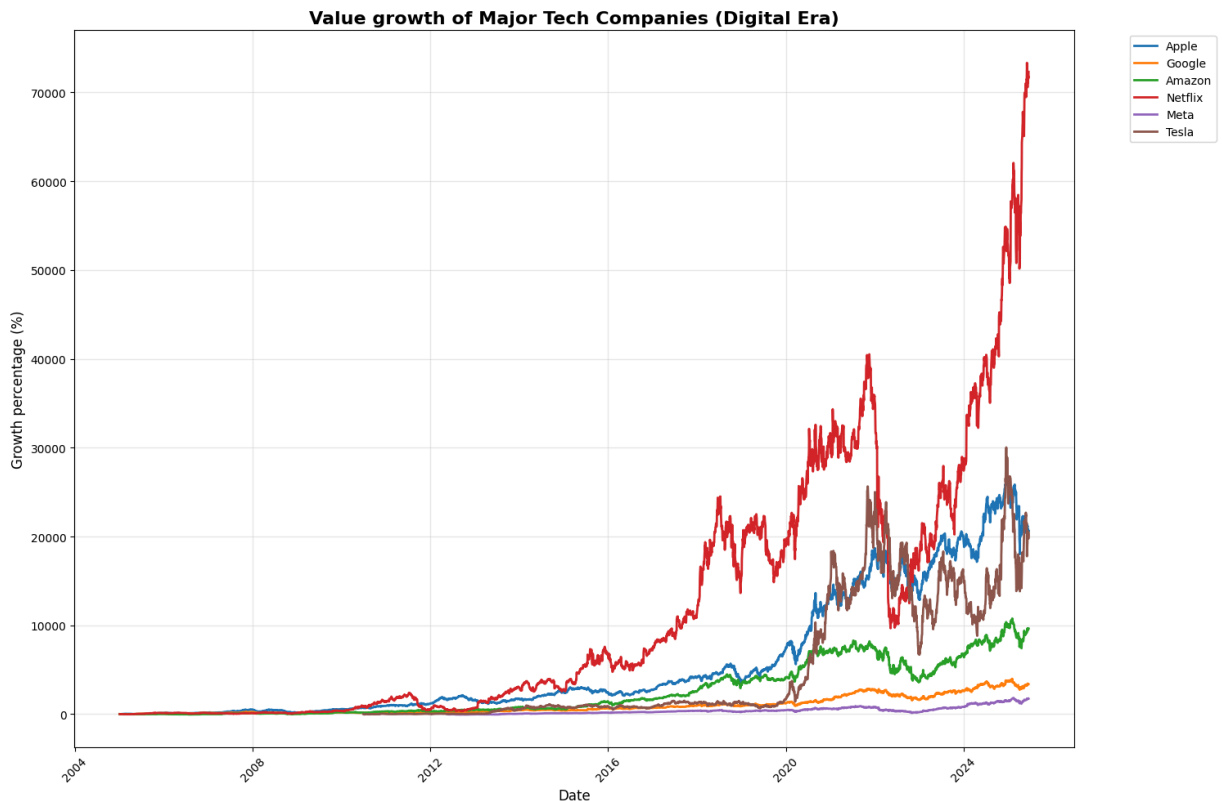
start_date = datetime(2005, 1, 1)

stockData = {}
for symbol, name in techStocks.items():
    ticker = yf.Ticker(symbol)
    data = ticker.history(start=start_date, end=end_date)
    stockData[name] = data['Close']

plt.figure(figsize=(15,10))
for name, prices in stockData.items():
    if len(prices) > 0:
        norm = (prices / prices.iloc[0] - 1) * 100
        plt.plot(norm.index, norm.values, linewidth=2, label=name)

plt.title('Value growth of Major Tech Companies (Digital Era)', fontsize=16, fontwe
plt.xlabel("Date", fontsize=12)
plt.ylabel("Growth percentage (%)", fontsize=12)
plt.legend(bbox_to_anchor=(1.05, 1), loc='upper left')
plt.grid(True, alpha=0.3)
plt.xticks(rotation=45)
plt.tight_layout()
plt.show()

```



```

In [17]: techStocks = {
    'AAPL': 'Apple',
    'GOOGL': 'Google',
    'AMZN': 'Amazon',
    'META': 'Meta',
    'TSLA': 'Tesla'
}

```

```

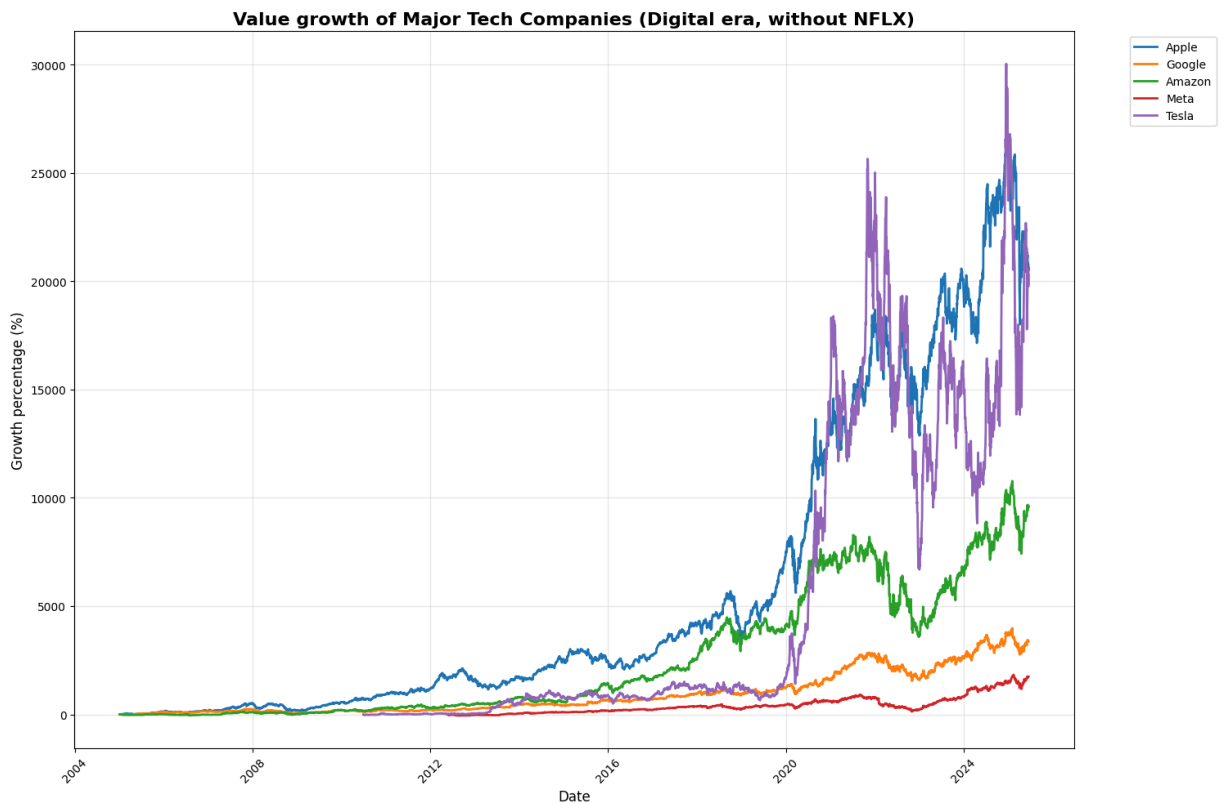
end_date = datetime.now()
start_date = datetime(2005, 1, 1)

stockData = {}
for symbol, name in techStocks.items():
    ticker = yf.Ticker(symbol)
    data = ticker.history(start=start_date, end=end_date)
    stockData[name] = data['Close']

plt.figure(figsize=(15,10))
for name, prices in stockData.items():
    if len(prices) > 0:
        norm = (prices / prices.iloc[0] - 1) * 100
        plt.plot(norm.index, norm.values, linewidth=2, label=name)

plt.title('Value growth of Major Tech Companies (Digital era, without NFLX)', fontsize=12)
plt.xlabel("Date", fontsize=12)
plt.ylabel("Growth percentage (%)", fontsize=12)
plt.legend(bbox_to_anchor=(1.05, 1), loc='upper left')
plt.grid(True, alpha=0.3)
plt.xticks(rotation=45)
plt.tight_layout()
plt.show()

```



```

In [16]: techStocks = {
    'NVDA': 'Nvidia',
    'PLTR': 'Palantir',
    'VST': 'Vistra',
    'NRG': 'NRG Energy',

```

```

'ORCL': 'Oracle'

}

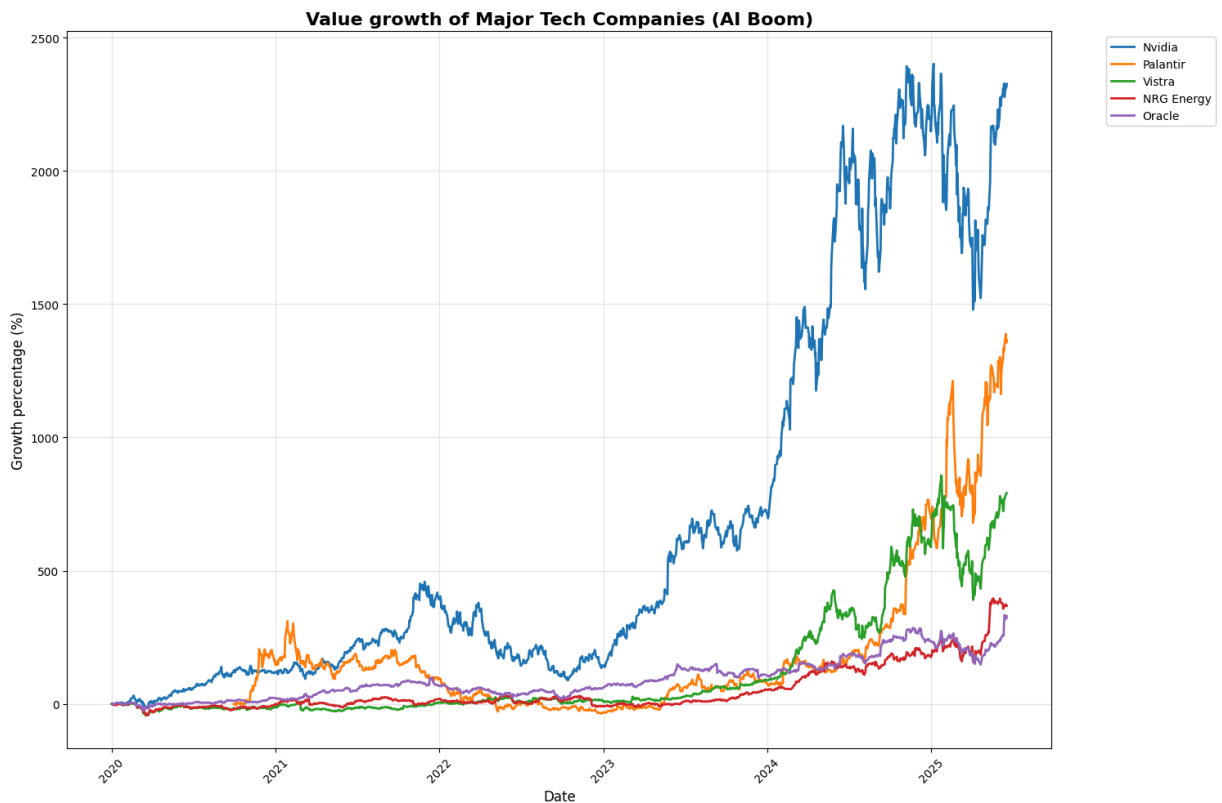
end_date = datetime.now()
start_date = datetime(2020, 1, 1)

stockData = {}
for symbol, name in techStocks.items():
    ticker = yf.Ticker(symbol)
    data = ticker.history(start=start_date, end=end_date)
    stockData[name] = data['Close']

plt.figure(figsize=(15,10))
for name, prices in stockData.items():
    if len(prices) > 0:
        norm = (prices / prices.iloc[0] - 1) * 100
        plt.plot(norm.index, norm.values, linewidth=2, label=name)

plt.title('Value growth of Major Tech Companies (AI Boom)', fontsize=16, fontweight
plt.xlabel("Date", fontsize=12)
plt.ylabel("Growth percentage (%)", fontsize=12)
plt.legend(bbox_to_anchor=(1.05, 1), loc='upper left')
plt.grid(True, alpha=0.3)
plt.xticks(rotation=45)
plt.tight_layout()
plt.show()

```



```

In [7]: techStocks = {
        'AAPL': 'Apple',

```

```

'BB': 'BlackBerry',
'SSNLF': 'Samsung',
'GOOG': 'Google',
'MSFT': 'Microsoft'
}

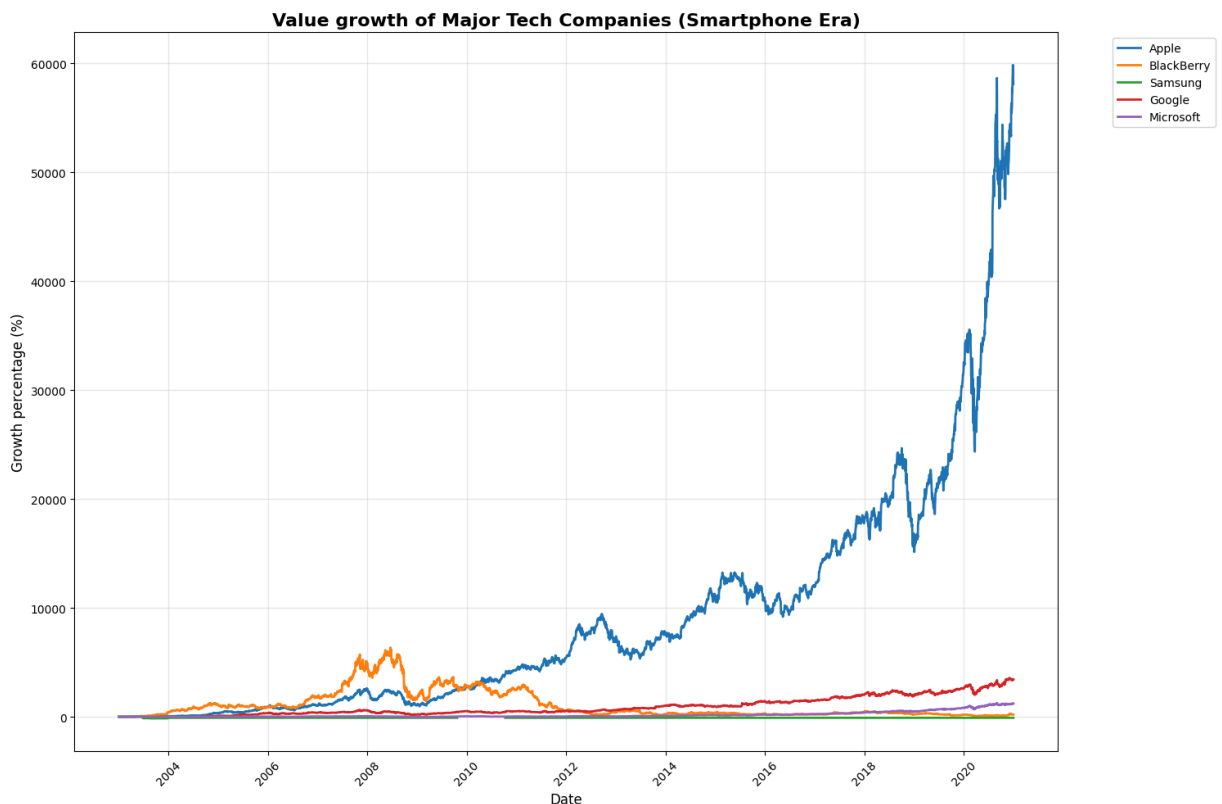
end_date = datetime(2021, 1, 1)
start_date = datetime(2003, 1, 1)

stockData = {}
for symbol, name in techStocks.items():
    ticker = yf.Ticker(symbol)
    data = ticker.history(start=start_date, end=end_date)
    stockData[name] = data['Close']

plt.figure(figsize=(15,10))
for name, prices in stockData.items():
    if len(prices) > 0:
        norm = (prices / prices.iloc[0] - 1) * 100
        plt.plot(norm.index, norm.values, linewidth=2, label=name)

plt.title('Value growth of Major Tech Companies (Smartphone Era)', fontsize=16, font
plt.xlabel("Date", fontsize=12)
plt.ylabel("Growth percentage (%)", fontsize=12)
plt.legend(bbox_to_anchor=(1.05, 1), loc='upper left')
plt.grid(True, alpha=0.3)
plt.xticks(rotation=45)
plt.tight_layout()
plt.show()

```



```
In [31]: techStocks = {
    'CSCO': 'Cisco Systems',
    'AMZN': 'Amazon',
    'EBAY': 'Ebay',
    'MSFT': 'Microsoft',
    'ORCL': 'Oracle',
    'INTC': 'Intel'

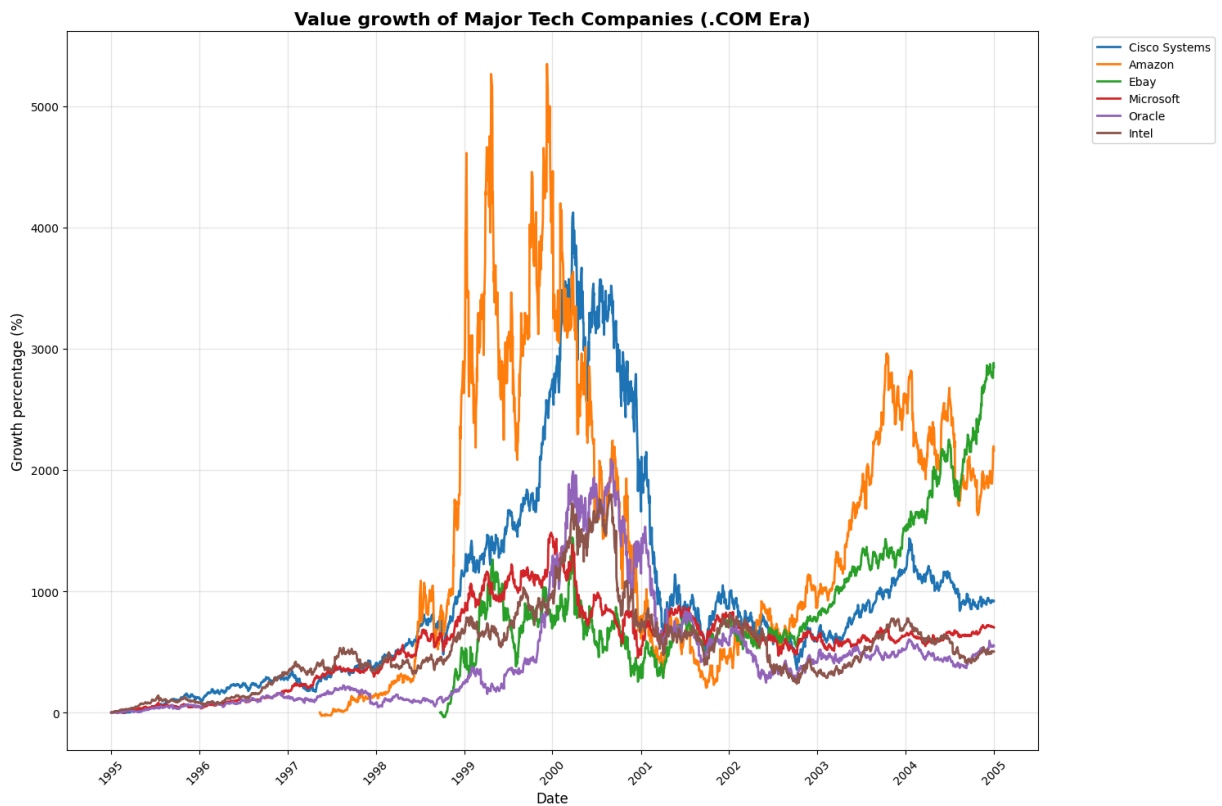
}

end_date = datetime(2005, 1, 1)
start_date = datetime(1995, 1, 1)

stockData = {}
for symbol, name in techStocks.items():
    ticker = yf.Ticker(symbol)
    data = ticker.history(start=start_date, end=end_date)
    stockData[name] = data['Close']

plt.figure(figsize=(15,10))
for name, prices in stockData.items():
    if len(prices) > 0:
        norm = (prices / prices.iloc[0] - 1) * 100
        plt.plot(norm.index, norm.values, linewidth=2, label=name)

plt.title('Value growth of Major Tech Companies (.COM Era)', fontsize=16, fontweight='bold')
plt.xlabel("Date", fontsize=12)
plt.ylabel("Growth percentage (%)", fontsize=12)
plt.legend(bbox_to_anchor=(1.05, 1), loc='upper left')
plt.grid(True, alpha=0.3)
plt.xticks(rotation=45)
plt.tight_layout()
plt.show()
```



```
In [36]: techStocks = {
    'IBM': 'IBM',
    'MSFT': 'Microsoft',
    'INTC': 'Intel',
    'AAPL': 'Apple',
    'HPQ': 'HP',
    'TXN': 'Texas Instruments'

}

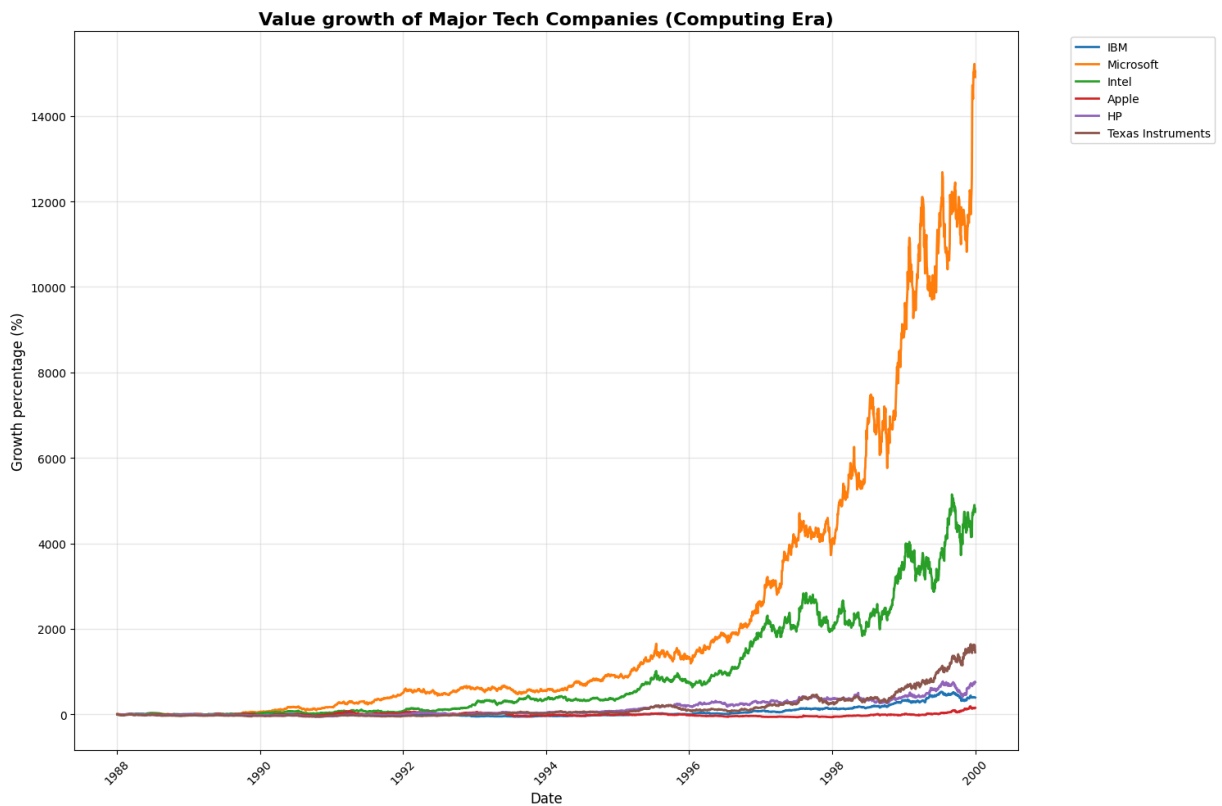
end_date = datetime(2000, 1, 1)
start_date = datetime(1988, 1, 1)

stockData = {}
for symbol, name in techStocks.items():
    ticker = yf.Ticker(symbol)
    data = ticker.history(start=start_date, end=end_date)
    stockData[name] = data['Close']

plt.figure(figsize=(15,10))
for name, prices in stockData.items():
    if len(prices) > 0:
        norm = (prices / prices.iloc[0] - 1) * 100
        plt.plot(norm.index, norm.values, linewidth=2, label=name)

plt.title('Value growth of Major Tech Companies (Computing Era)', fontsize=16, font
plt.xlabel("Date", fontsize=12)
plt.ylabel("Growth percentage (%)", fontsize=12)
```

```
plt.legend(bbox_to_anchor=(1.05, 1), loc='upper left')
plt.grid(True, alpha=0.3)
plt.xticks(rotation=45)
plt.tight_layout()
plt.show()
```



```
In [37]: techStocks = {
    'IBM': 'IBM',
    'AAPL': 'Apple',
    'HPQ': 'HP',
    'TXN': 'Texas Instruments'
}

end_date = datetime(2000, 1, 1)
start_date = datetime(1988, 1, 1)

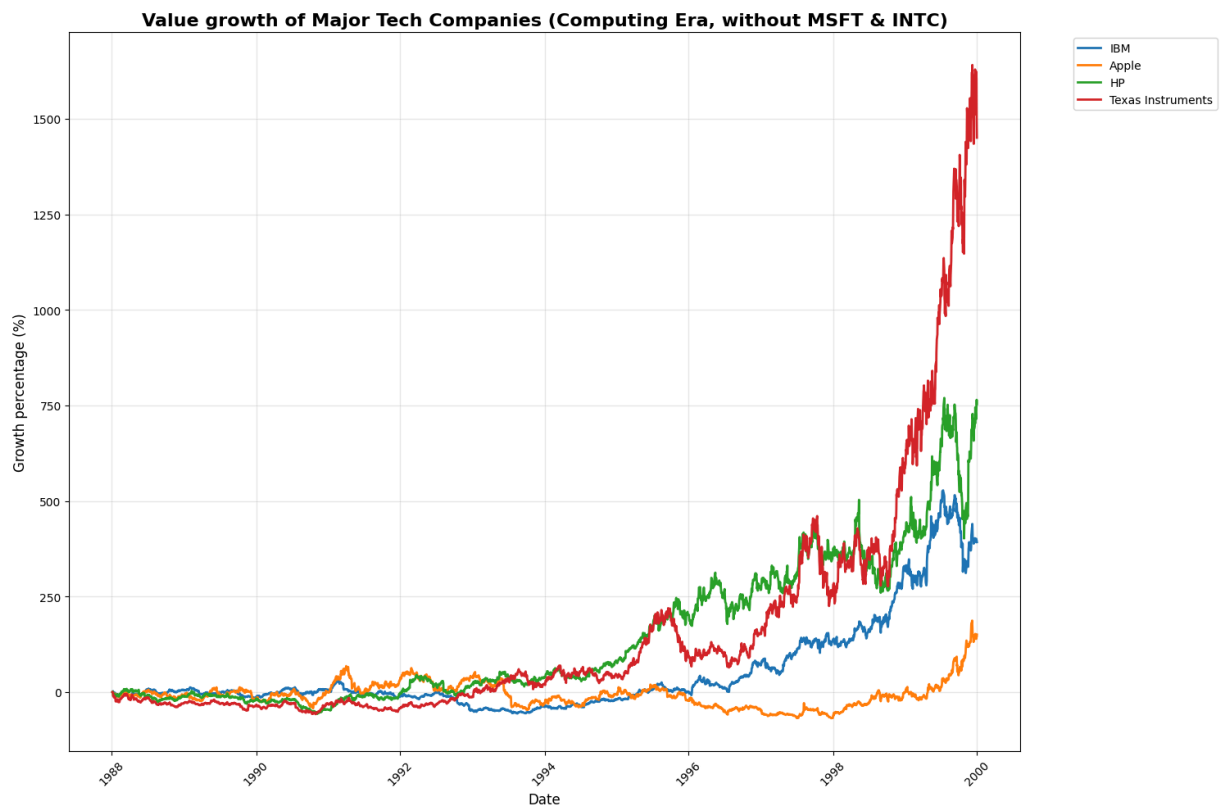
stockData = {}
for symbol, name in techStocks.items():
    ticker = yf.Ticker(symbol)
    data = ticker.history(start=start_date, end=end_date)
    stockData[name] = data['Close']

plt.figure(figsize=(15,10))
for name, prices in stockData.items():
    if len(prices) > 0:
        norm = (prices / prices.iloc[0] - 1) * 100
        plt.plot(norm.index, norm.values, linewidth=2, label=name)
```

```

plt.title('Value growth of Major Tech Companies (Computing Era, without MSFT & INTC)
plt.xlabel("Date", fontsize=12)
plt.ylabel("Growth percentage (%)", fontsize=12)
plt.legend(bbox_to_anchor=(1.05, 1), loc='upper left')
plt.grid(True, alpha=0.3)
plt.xticks(rotation=45)
plt.tight_layout()
plt.show()

```



```

In [3]: techStocks = {
    'AAPL': 'Apple',
    'PLTR': 'Palantir',
    'GOOGL': 'Google',
    'AMZN': 'Amazon',
    'TSLA': 'Tesla',
    'NVDA': 'NVIDIA',
    'NFLX': 'Netflix',
    'META': 'Meta',
    'MSFT': 'Microsoft'
}

end_date = datetime.now()
start_date = end_date - timedelta(days=1825)

stockData = {}
for symbol, name in techStocks.items():
    ticker = yf.Ticker(symbol)
    data = ticker.history(start=start_date, end=end_date)
    shareCount = ticker.info.get('sharesOutstanding')
    marketVal = data['Close'] * shareCount

```



```

stockData[name] = marketVal

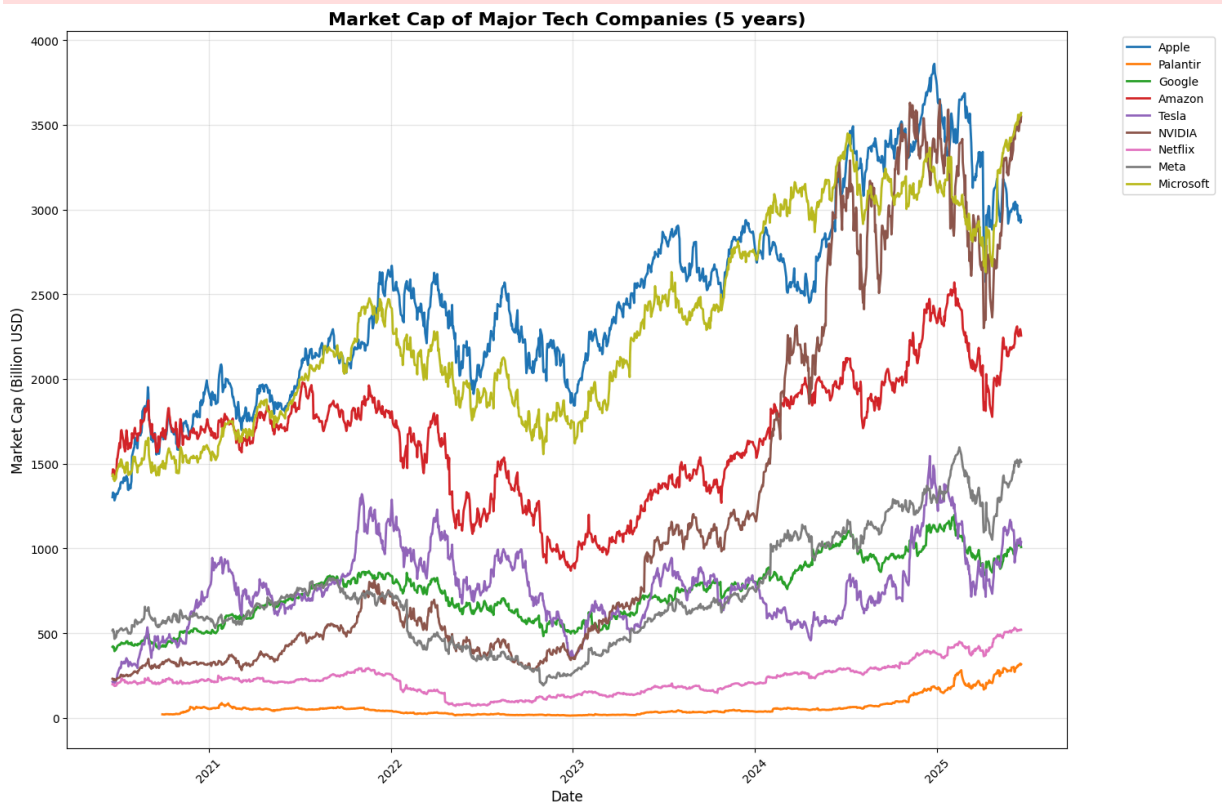
plt.figure(figsize=(15,10))
for name, prices in stockData.items():
    if len(prices) > 0:
        plt.plot(prices.index, prices.values / 1e9, linewidth=2, label=name)

plt.title('Market Cap of Major Tech Companies (5 years)', fontsize=16, fontweight='bold')
plt.xlabel("Date", fontsize=12)
plt.ylabel("Market Cap (Billion USD)", fontsize=12)
plt.legend(bbox_to_anchor=(1.05, 1), loc='upper left')
plt.grid(True, alpha=0.3)
plt.xticks(rotation=45)
plt.tight_layout()
plt.show()

print("\nEstimated Current Market Capitalizations:")
print("-" * 50)
for symbol, name in techStocks.items():
    ticker = yf.Ticker(symbol)
    currentPrice = ticker.history(period="1d")['Close'].iloc[-1]
    shares = ticker.info.get('sharesOutstanding')
    marketCap = currentPrice * shares
    print(f"{name:12} ({symbol}): ${marketCap / 1e9:.2f} Billion")

```

HTTP Error 401:



Estimated Current Market Capitalizations:

Apple	(AAPL): \$2936.08 Billion
Palantir	(PLTR): \$316.72 Billion
Google	(GOOGL): \$1008.72 Billion
Amazon	(AMZN): \$2256.20 Billion

HTTP Error 401:

Tesla	(TSLA): \$1037.31 Billion
NVIDIA	(NVDA): \$3547.91 Billion
Netflix	(NFLX): \$520.17 Billion
Meta	(META): \$1510.62 Billion

HTTP Error 401:

Microsoft	(MSFT): \$3569.40 Billion
-----------	---------------------------

```
In [4]: techStocks = {
    'AAPL': 'Apple',
    'GOOGL': 'Google',
    'AMZN': 'Amazon',
    'TSLA': 'Tesla',
    'NVDA': 'NVIDIA',
    'NFLX': 'Netflix',
    'META': 'Meta',
    'MSFT': 'Microsoft'
}

end_date = datetime.now()
start_date = end_date - timedelta(days=3650)

stockData = {}
for symbol, name in techStocks.items():
    ticker = yf.Ticker(symbol)
    data = ticker.history(start=start_date, end=end_date)
    shareCount = ticker.info.get('sharesOutstanding')
    marketVal = data['Close'] * shareCount
    stockData[name] = marketVal

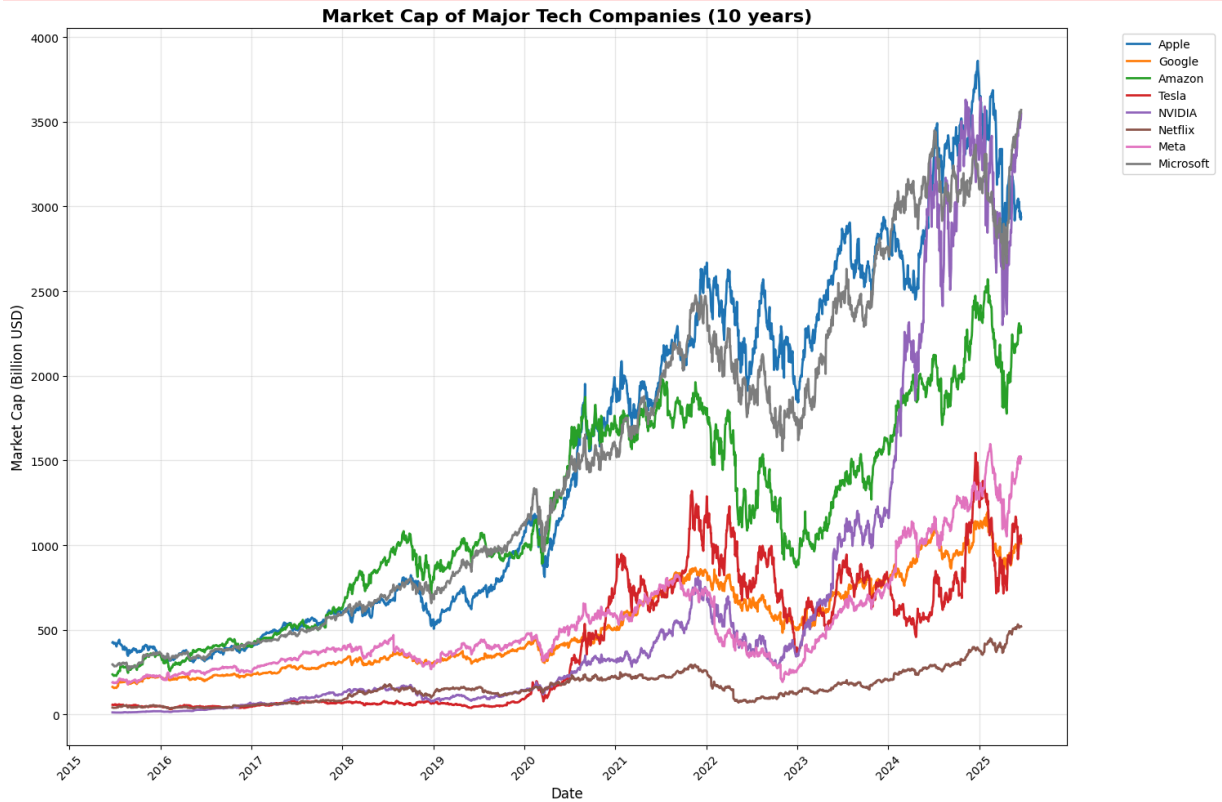
plt.figure(figsize=(15,10))
for name, prices in stockData.items():
    if len(prices) > 0:
        plt.plot(prices.index, prices.values / 1e9, linewidth=2, label=name)

plt.title('Market Cap of Major Tech Companies (10 years)', fontsize=16, fontweight=
plt.xlabel("Date", fontsize=12)
plt.ylabel("Market Cap (Billion USD)", fontsize=12)
plt.legend(bbox_to_anchor=(1.05, 1), loc='upper left')
plt.grid(True, alpha=0.3)
plt.xticks(rotation=45)
plt.tight_layout()
plt.show()

print("\nEstimated Current Market Capitalizations:")
print("-" * 50)
for symbol, name in techStocks.items():
    ticker = yf.Ticker(symbol)
```

```
currentPrice = ticker.history(period="1d")['Close'].iloc[-1]
shares = ticker.info.get('sharesOutstanding')
marketCap = currentPrice * shares
print(f"{name:12} ({symbol}): ${marketCap / 1e9:.2f} Billion")
```

HTTP Error 401:



Estimated Current Market Capitalizations:

HTTP Error 401:

Apple	(AAPL): \$2936.08 Billion
Google	(GOOGL): \$1008.72 Billion
Amazon	(AMZN): \$2256.20 Billion
Tesla	(TSLA): \$1037.31 Billion
NVIDIA	(NVDA): \$3547.91 Billion
Netflix	(NFLX): \$520.17 Billion
Meta	(META): \$1510.62 Billion
Microsoft	(MSFT): \$3569.40 Billion

```
In [8]: techStocks = {
    'AAPL': 'Apple',
    'GOOGL': 'Google',
    'AMZN': 'Amazon',
    'TSLA': 'Tesla',
    'NVDA': 'NVIDIA',
    'NFLX': 'Netflix',
    'META': 'Meta',
    'MSFT': 'Microsoft'
}

end_date = datetime.now()
start_date = end_date - timedelta(days=10950)
```

```

stockData = {}
for symbol, name in techStocks.items():
    ticker = yf.Ticker(symbol)
    data = ticker.history(start=start_date, end=end_date)
    shareCount = ticker.info.get('sharesOutstanding')
    marketVal = data['Close'] * shareCount
    stockData[name] = marketVal

plt.figure(figsize=(15,10))
for name, prices in stockData.items():
    if len(prices) > 0:
        plt.plot(prices.index, prices.values / 1e9, linewidth=2, label=name)

plt.title('Market Cap of Major Tech Companies (30 years)', fontsize=16, fontweight=
plt.xlabel("Date", fontsize=12)
plt.ylabel("Market Cap (Billion USD)", fontsize=12)
plt.legend(bbox_to_anchor=(1.05, 1), loc='upper left')
plt.grid(True, alpha=0.3)
plt.xticks(rotation=45)
plt.tight_layout()
plt.show()

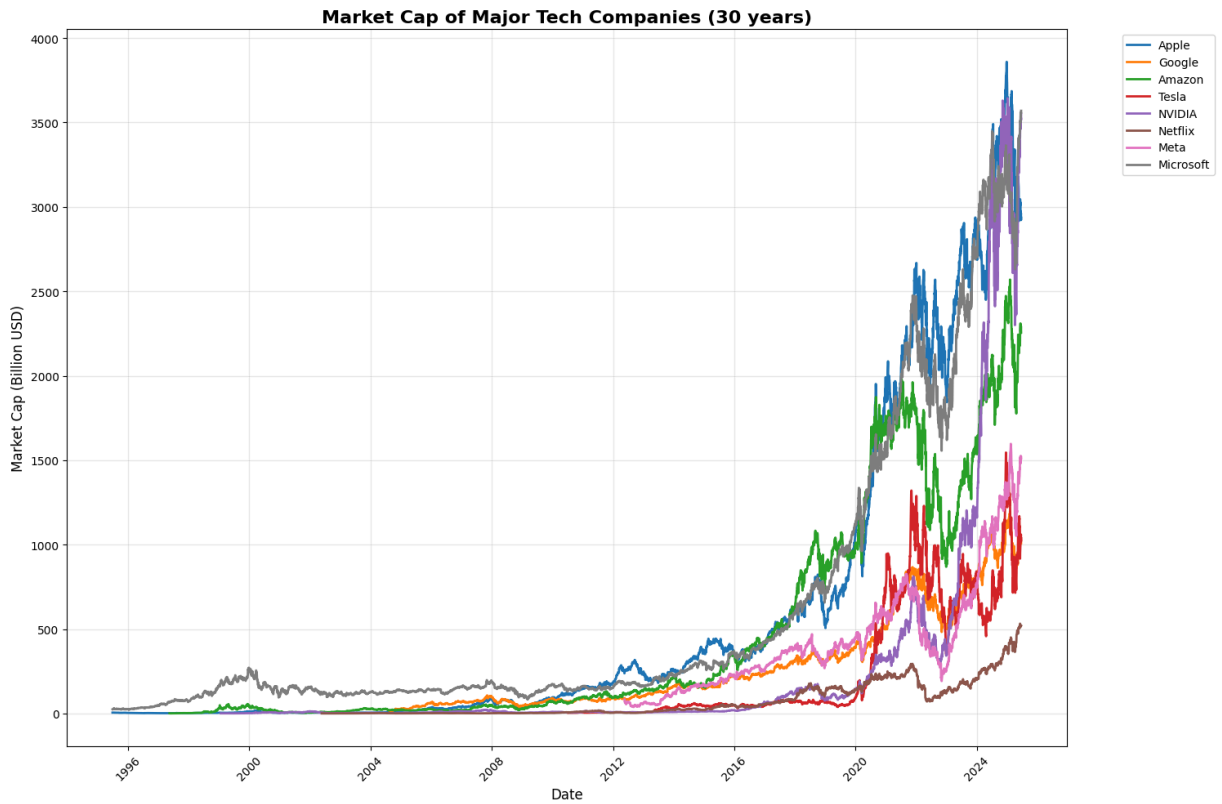
print("\nEstimated Current Market Capitalizations:")
print("-" * 50)
for symbol, name in techStocks.items():
    ticker = yf.Ticker(symbol)
    currentPrice = ticker.history(period="1d")['Close'].iloc[-1]
    shares = ticker.info.get('sharesOutstanding')
    marketCap = currentPrice * shares
    print(f"{name:12} ({symbol}): ${marketCap / 1e9:.2f} Billion")

```

HTTP Error 401:

HTTP Error 401:

HTTP Error 401:



Estimated Current Market Capitalizations:

Apple	(AAPL): \$2936.08 Billion
Google	(GOOGL): \$1008.72 Billion
Amazon	(AMZN): \$2256.20 Billion
Tesla	(TSLA): \$1037.31 Billion
NVIDIA	(NVDA): \$3547.91 Billion
Netflix	(NFLX): \$520.17 Billion
Meta	(META): \$1510.62 Billion
Microsoft	(MSFT): \$3569.40 Billion

```
In [12]: techStocks = {
    'AAPL': 'Apple',
    'NVDA': 'NVIDIA',
    'MSFT': 'Microsoft'
}

end_date = datetime.now()
start_date = datetime(1995, 1, 1)

stockData = {}
for symbol, name in techStocks.items():
    ticker = yf.Ticker(symbol)
    data = ticker.history(start=start_date, end=end_date)
    shareCount = ticker.info.get('sharesOutstanding')
    marketVal = data['Close'] * shareCount
    stockData[name] = marketVal

plt.figure(figsize=(15,10))
for name, prices in stockData.items():
    if len(prices) > 0:
```

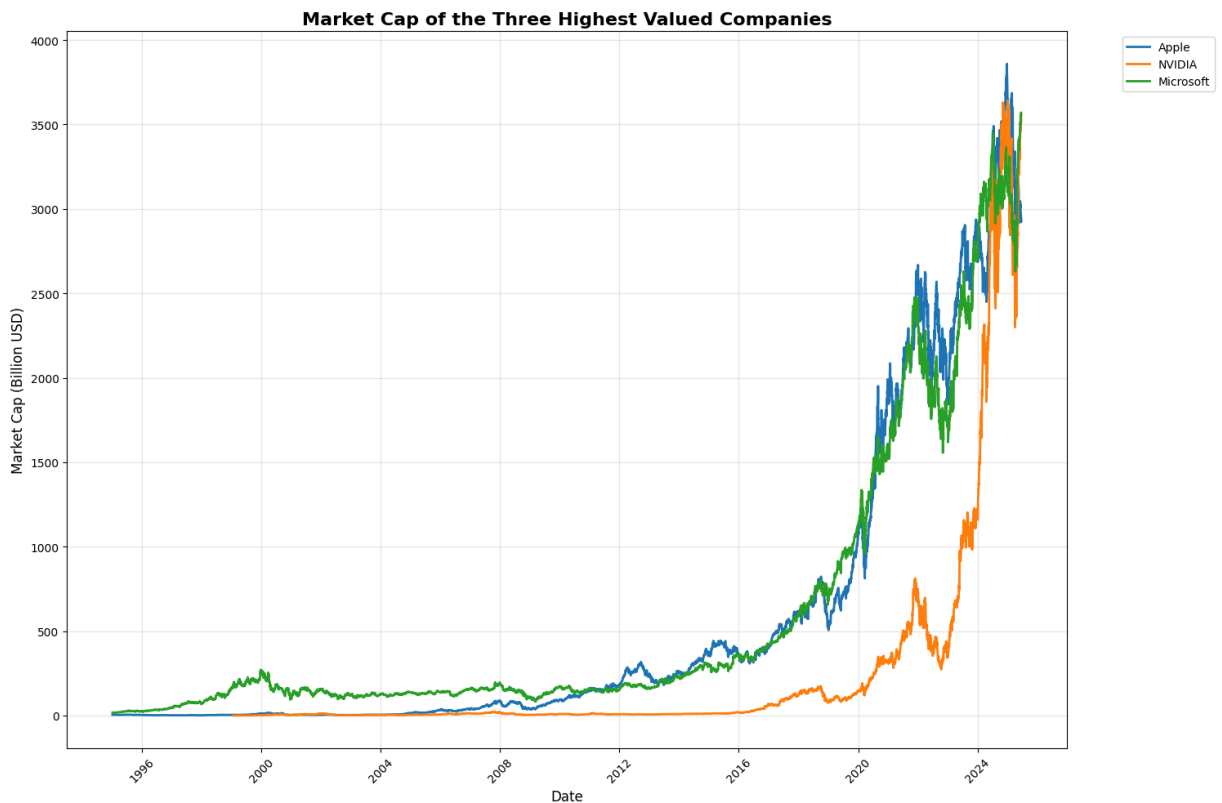
```

plt.plot(prices.index, prices.values / 1e9, linewidth=2, label=name)

plt.title('Market Cap of the Three Highest Valued Companies', fontsize=16, fontweig
plt.xlabel("Date", fontsize=12)
plt.ylabel("Market Cap (Billion USD)", fontsize=12)
plt.legend(bbox_to_anchor=(1.05, 1), loc='upper left')
plt.grid(True, alpha=0.3)
plt.xticks(rotation=45)
plt.tight_layout()
plt.show()

print("\nEstimated Current Market Capitalizations:")
print("-" * 50)
for symbol, name in techStocks.items():
    ticker = yf.Ticker(symbol)
    currentPrice = ticker.history(period="1d")['Close'].iloc[-1]
    shares = ticker.info.get('sharesOutstanding')
    marketCap = currentPrice * shares
    print(f"{name:12} ({symbol}): ${marketCap / 1e9:.2f} Billion")

```



Estimated Current Market Capitalizations:

```

-----
Apple      (AAPL): $2936.08 Billion
NVIDIA     (NVDA): $3547.91 Billion
Microsoft  (MSFT): $3569.40 Billion

```

```

In [4]: etf_tickers = {
        "IXN (MSFT, AAPL, NVDA)": "IXN",
        "IYW (MSFT, NVDA, AAPL)": "IYW",
        "VGT (AAPL, MSFT, NVDA)": "VGT",
        "XLK (MSFT, NVDA, AAPL)": "XLK",
        "SMH (NVDA, Semiconductors)": "SMH",

```

```

    "QQQ (MSFT, NVDA, AAPL)": "QQQ",
    "Nasdaq": "NDAQ"

}

market_ticker = "^GSPC"

start_date = "1990-01-01"
end_date = "2025-06-20"

tickers = list(etf_tickers.values()) + [market_ticker]

data = yf.download(
    tickers,
    start=start_date,
    end=end_date,
    progress=False,
    auto_adjust=False
)["Adj Close"]

data = data.rename(columns={v: k for k, v in etf_tickers.items()})

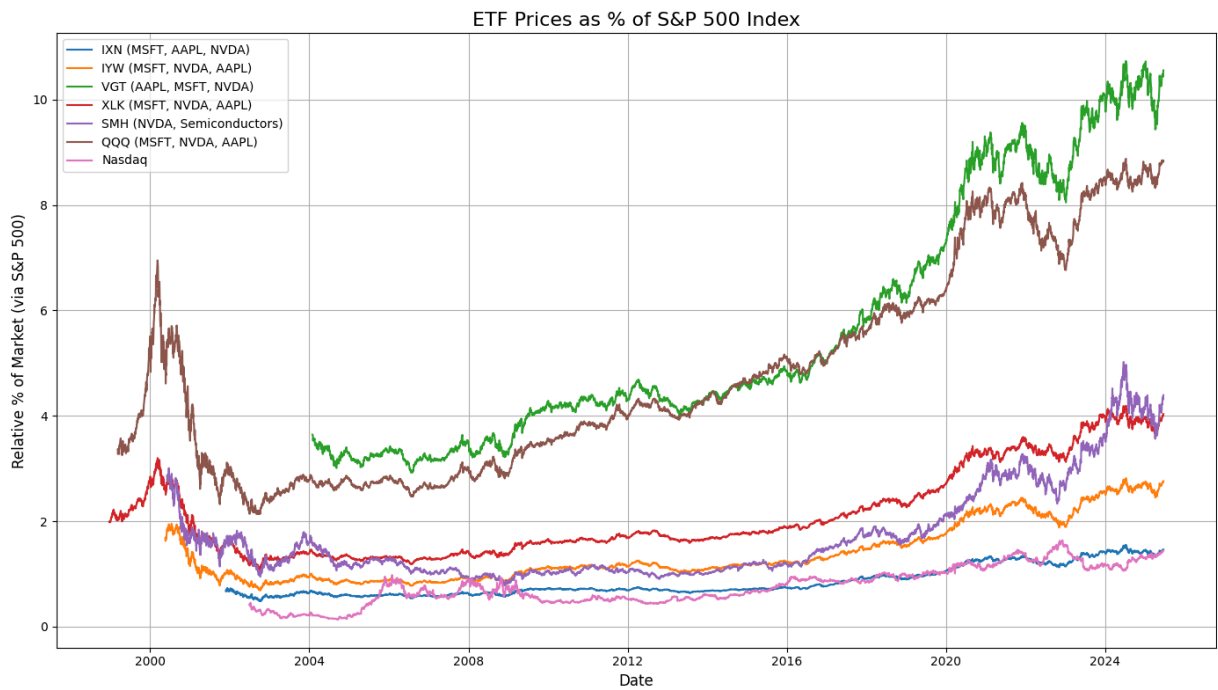
market_col = "^GSPC"

percent_of_market = data[list(etf_tickers.keys())].div(data[market_col], axis=0) *

plt.figure(figsize=(14, 8))
for col in percent_of_market.columns:
    plt.plot(percent_of_market.index, percent_of_market[col], label=col)

plt.title("ETF Prices as % of S&P 500 Index", fontsize=16)
plt.xlabel("Date", fontsize=12)
plt.ylabel("Relative % of Market (via S&P 500)", fontsize=12)
plt.legend()
plt.grid(True)
plt.tight_layout()
plt.show()

```



```
In [6]: etf_tickers = {
    "IXN (MSFT, AAPL, NVDA)": "IXN",
    "IYW (MSFT, NVDA, AAPL)": "IYW",
    "VGT (AAPL, MSFT, NVDA)": "VGT",
    "XLK (MSFT, NVDA, AAPL)": "XLK",
    "SMH (NVDA, Semiconductors)": "SMH",
    "QQQ (MSFT, NVDA, AAPL)": "QQQ",
    "Nasdaq": "NDAQ"
}

market_ticker = "^GSPC"

start_date = "1998-01-01"
end_date = "2025-06-20"

tickers = list(etf_tickers.values()) + [market_ticker]

data = yf.download(
    tickers,
    start=start_date,
    end=end_date,
    progress=False,
    auto_adjust=False
)["Adj Close"]

data = data.rename(columns={v: k for k, v in etf_tickers.items()})
data[market_ticker] = data[market_ticker].ffill()

relative_performance = pd.DataFrame(index=data.index)

for etf_name in etf_tickers.keys():
    etf_series = data[etf_name].dropna()
    sp500_series = data[market_ticker].loc[etf_series.index]
```



```

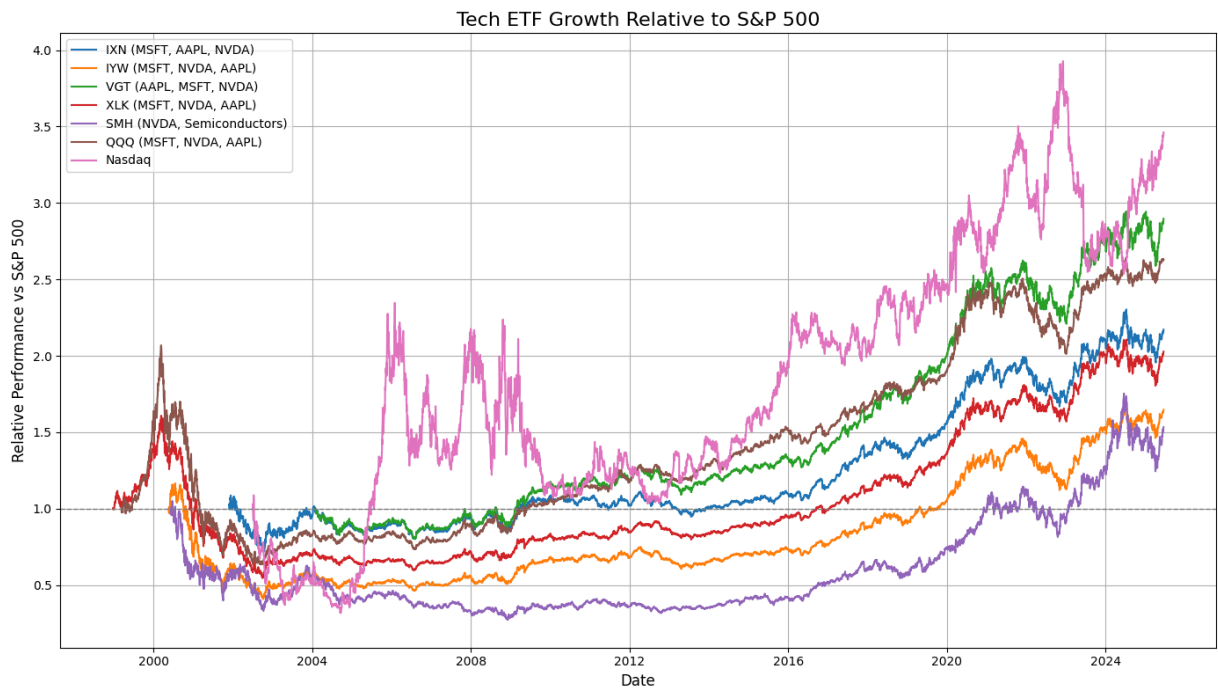
etf_norm = etf_series / etf_series.iloc[0]
sp500_norm = sp500_series / sp500_series.iloc[0]

relative = etf_norm / sp500_norm
relative_performance[etf_name] = relative

plt.figure(figsize=(14, 8))
for col in relative_performance.columns:
    plt.plot(relative_performance.index, relative_performance[col], label=col)

plt.title("Tech ETF Growth Relative to S&P 500", fontsize=16)
plt.xlabel("Date", fontsize=12)
plt.ylabel("Relative Performance vs S&P 500", fontsize=12)
plt.axhline(1, color='gray', linestyle='--', linewidth=1)
plt.legend()
plt.grid(True)
plt.tight_layout()
plt.show()

```



```

In [5]: etf_tickers = {
    "IXN (MSFT, AAPL, NVDA)": "IXN",
    "IYW (MSFT, NVDA, AAPL)": "IYW",
    "VGT (AAPL, MSFT, NVDA)": "VGT",
    "XLK (MSFT, NVDA, AAPL)": "XLK",
    "SMH (NVDA, Semiconductors)": "SMH",
    "QQQ (MSFT, NVDA, AAPL)": "QQQ",
    "Nasdaq": "NDAQ"
}

```

```
market_ticker = "^GSPC"
```

```
start_date = "2003-01-01"
```

```
end_date = "2025-06-20"
```

```

tickers = list(etf_tickers.values()) + [market_ticker]

data = yf.download(
    tickers,
    start=start_date,
    end=end_date,
    progress=False,
    auto_adjust=False
)["Adj Close"]

data = data.rename(columns={v: k for k, v in etf_tickers.items()})
data[market_ticker] = data[market_ticker].ffill()

relative_performance = pd.DataFrame(index=data.index)

for etf_name in etf_tickers.keys():
    etf_series = data[etf_name].dropna()
    sp500_series = data[market_ticker].loc[etf_series.index]

    etf_norm = etf_series / etf_series.iloc[0]
    sp500_norm = sp500_series / sp500_series.iloc[0]

    relative = etf_norm / sp500_norm
    relative_performance[etf_name] = relative

plt.figure(figsize=(14, 8))
for col in relative_performance.columns:
    plt.plot(relative_performance.index, relative_performance[col], label=col)

plt.title("Tech ETF Growth Relative to S&P 500 (Since Burst)", fontsize=16)
plt.xlabel("Date", fontsize=12)
plt.ylabel("Relative Performance vs S&P 500", fontsize=12)
plt.axhline(1, color='gray', linestyle='--', linewidth=1)
plt.legend()
plt.grid(True)
plt.tight_layout()
plt.show()

```

